

Physics-augmented neural networks and projection-based reduced-order models for anisotropic hyperelasticity

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Abstract

The repeated representative volume element (RVE) solves required by finite-strain FE^2 calculations motivate surrogates that reduce their online computational cost. This work compares non-intrusive physics-augmented neural networks (PANNs) and intrusive projection-based reduced-order models (PROMs) for anisotropic hyperelasticity. For the PANNs, paired directional features with trainable orientations, powers, and determinant exponents are coupled to either an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). The features preserve objectivity and two-dimensional polyconvexity while admitting an isotropic reference derivative, so that affine determinant normalization separates the stress-free reference condition from volumetric stiffness. A separate sufficient lower bound addresses nonnegative energy for the selected parameters. The PROM-based alternatives comprise a linear ECM-based HPROM, an HPROM–ANN with primary/secondary latent-space closure and adaptive cubature, and a direct decoder variant without microscopic equilibrium iterations. All use genuine reduced meshes. A periodic cell containing a rotated elliptical void tests approximation and mechanical consistency: the selected polyconvex models attain approximately 0.65% held-out stress error, while a closed strain cycle reveals a nonconservative defect in direct stress regression. A common tensile-coupon calculation then measures structural error and online cost against nested $\text{FOM}\text{--}\text{FE}^2$. The iterative HPROM–ANN reduces the microscopic equilibrium dimension from 39 to three and is 2.23 times faster than the optimized linear HPROM, with a 0.052% displacement error. ICNN and ICKAN yield displacement errors below 0.43% in 1.73 and 3.67 seconds under the reported thread configurations. These results establish distinct accuracy–cost operating points and demonstrate that accurate polyconvex learning is compatible with this anisotropic homogenized response. Constitutive guarantees, sampled extrapolation behavior, and nominal finite element convergence are assessed separately.

Keywords: computational homogenization; hyperelasticity; physics-augmented neural networks; polyconvexity; projection-based reduced-order models; empirical cubature.

1 Introduction

Computational homogenization provides a way to predict the response of heterogeneous materials without prescribing a macroscopic constitutive equation in advance. Instead, the effective stress and tangent are obtained by solving a boundary-value problem on a representative volume element (RVE) [1]. In a two-level finite element calculation, commonly denoted FE^2 , this problem is coupled to the macroscopic finite element model: the deformation at each macroscopic integration point drives an RVE calculation, whose homogenized response enters the structural equilibrium equations. This allows the influence of the constituents and microstructure to be carried into a structural analysis, including anisotropy and coupling between deformation modes at finite strains.

The computational burden is substantial. Each macroscopic load increment can require several Newton iterations, and each iteration entails constitutive evaluations at many integration points. In FE^2 , these evaluations are themselves nonlinear finite element solves. Their repetition, together with the storage and manipulation of microscopic states and operators, can make the overall calculation prohibitively demanding in execution time and computational resources. This motivates surrogate models that replace the repeated

full-order RVE calculations with less expensive evaluations. Representative full-order solutions are generated offline; the objective is then to exploit this information to reduce the online cost while retaining the accuracy needed by the macroscopic solver.

Two broad routes can be distinguished by their access to the microscopic governing equations. Non-intrusive constitutive surrogates learn the relation between macroscopic deformation and homogenized response from RVE data, without requiring access to the microscopic residual and tangent operators. Once trained, they replace the online RVE solve with a direct evaluation of stress and tangent. Intrusive projection-based reduced-order models (PROMs), in contrast, use those operators to construct and solve a low-dimensional approximation of the microscopic equilibrium problem. Hyperreduction further limits the elements or integration points needed to evaluate the reduced equations. Both routes seek computational savings, but they replace different parts of the full-order calculation.

This distinction is not a division between models with and without physics, nor between models with and without neural networks. A non-intrusive surrogate can embed constitutive restrictions in its architecture, as in physics-augmented neural networks (PANNs), while a PROM can use an artificial neural network (ANN) to enrich its approximation space and still retain a reduced equilibrium solve. The questions are therefore what the surrogate learns, which equations or constitutive properties it retains, and how these choices affect structural accuracy and online cost. These questions motivate the present comparison of PANNs and PROM-based surrogates for anisotropic hyperelasticity.

1.1 Physics-augmented neural networks

The non-intrusive route begins with a constitutive approximation: given the macroscopic deformation, predict the response that would otherwise require an RVE solve. Neural models of homogenized material behavior [2] and physics-constrained networks coupled to automated microscopic data generation [3] illustrate this approach. Its attraction is that the online model no longer needs a microscopic mesh or equilibrium iterations. Its central difficulty is ensuring that the learned relation remains a suitable constitutive law when used inside a structural solver. The required state description also depends on the material: learning microscopic histories [4], directional polymer models with history variables [5], and staggered training across interacting scales in woven materials [6] address path-dependent behavior. The present work concerns history-independent hyperelasticity, for which the current deformation describes the constitutive state.

Even in this setting, fitting stress values is not sufficient to define a mechanically consistent constitutive law. For a hyperelastic material, the stress must derive from a stored-energy function. Direct regression of its components does not enforce that relation: an accurate fit at sampled states can coexist with nonzero net work around a closed deformation cycle. Learning a scalar energy and obtaining stress by differentiation enforces the potential structure, as emphasized by As’ad et al. [7], and supplies the corresponding major symmetry of the tangent. Objectivity can also be imposed through deformation measures that leave the energy unchanged under superposed rigid rotations. PANNs incorporate such constitutive structure into the learned representation [8, 9]. The term identifies this modeling strategy, not a single architecture or a guarantee that every PANN satisfies the same set of conditions. Which properties hold must be established for the particular construction.

Curvature introduces a further distinction. An energy potential need not satisfy a convexity condition, and a convexity condition must specify its variables. As’ad et al. [7] use an input-convex neural network (ICNN) in Green–Lagrange strain. Convexity in that strain measure has a tensile-state stability implication, but the second Piola stress enters the relation between strain-space and deformation-gradient curvature [10]. Similarly, a positive-matrix factorization in an incremental stress update [11] is not, by itself, an energy Hessian. These formulations therefore address different constraints.

Polyconvexity [12] provides a tractable sufficient condition for rank-one convexity by representing the energy as a convex function of deformation minors. In two dimensions, a convex representation in the deformation gradient and its determinant is sufficient. Together with appropriate growth, coercivity, and admissibility hypotheses, this structure also supports existence results in nonlinear elasticity [13]. It does not guarantee uniqueness or convergence of a particular Newton algorithm. For anisotropic hyperelasticity, the challenge is to impose the selected conditions without making the representation too restrictive to fit the homogenized response. This is the constitutive question pursued here. The alternative PROM route approaches the acceleration problem from the governing equations rather than from a prescribed macroscopic energy representation.

1.2 Projection-based reduced-order models

The intrusive route retains the governing equilibrium equations instead of learning only their homogenized constitutive output. Projection-based model order reduction (PMOR) seeks their solution in a low-

dimensional approximation space. For a microscopic displacement problem, a conventional PROM represents the displacement field in an affine space spanned by a reduced-order basis, usually constructed by proper orthogonal decomposition (POD) of solution snapshots. The basis coefficients are the generalized coordinates solved for online. Reduced finite-strain homogenization [14, 15] and potential-based two-scale calculations with GPU acceleration [16] establish this route to reducing microscopic cost. Reduced periodic-substructure methods also operate in domain-decomposition settings [17]; here, the reduced model serves as the microscopic evaluator within first-order FE².

There are two distinct costs to reduce. The first is the number of independent unknowns. A small affine space may represent a curved family of microscopic solutions poorly, even when that family depends on few loading parameters. Local reduced bases [18], autoencoder-based manifolds [19], and quadratic approximation manifolds [20] address this representational difficulty in different ways. The PROM-ANN approach of Barnett et al. [21] and the latent-space closure framework of Ares De Parga et al. [22] instead partition the reduced coordinates into primary and secondary groups. The primary generalized coordinates remain independent unknowns, while a learned closure predicts the secondary generalized coordinates from them. A rich displacement representation can therefore be reconstructed while solving for only a few independent coordinates. We adopt this primary/secondary nomenclature.

The second cost is the evaluation of the reduced equations. Reducing the number of coordinates is insufficient if every residual evaluation still requires constitutive calculations and assembly over the full microscopic mesh. Sampling-and-weighting hyperreduction approximates these evaluations through a selected subset of local contributions and associated weights. Optimized cubature [23], energy-conserving sampling and weighting (ECSW) [24, 25], and the empirical cubature method (ECM) [26] provide established constructions, including extensions to local bases and Petrov-Galerkin projection [27, 28]. A hyperreduced projection-based reduced-order model (HPROM) thus combines an approximation of the microscopic state with an approximation of the integrals entering its reduced equations.

The integration rule can itself exploit the state representation. Continuous empirical cubature adjusts point locations and weights [29]; subspace-adaptive cubature uses common locations with weights specific to each reduced subspace [30]. Manifold-adaptive weights extend the latter idea to weights varying continuously with the reduced coordinates [31]. We use fixed-weight ECM for the affine HPROM and manifold-adaptive cubature for HPROM-ANN, both on actual reduced meshes. These models retain a reduced equilibrium solve. A direct operating mode, D-HPROM-ANN, additionally uses the imposed strain to prescribe strain-informed coordinates and reconstruct the microscopic response without equilibrium iterations. Although this mode also avoids an online equilibrium solve, it is derived from the intrusive reduced-order construction and still evaluates the microscopic material law on selected elements. It therefore differs from a PANN that directly represents the homogenized energy.

1.3 Feature representation and position relative to prior work

The comparison between these routes brings the representational question into focus. A PROM approximates the microscopic displacement field and evaluates the given constituent law, whereas an energy-based PANN must represent the effective anisotropic energy itself. For the latter, the choice of constitutive inputs is part of the model. Structural tensors encode material directions, and invariants formed from these tensors provide objective measures of deformation. Polyconvex invariant formulations have been developed for anisotropic elasticity [32], while models of distributed fiber orientations illustrate why the directional description can affect predicted deformation mechanisms [33]. In a neural constitutive model, these scalar measures act as the network’s features.

A useful sufficient construction combines polyconvex features with an energy core that is convex and nondecreasing in every feature. ICNNs provide the convex core [34]; anisotropic invariant formulations and compatible growth and reference-state corrections are developed by Klein et al. [8] and Linden et al. [9]. Other structured parameterizations include constitutive artificial neural networks [35] and polyconvex neural ordinary differential equations [36]. For spline-based Kolmogorov-Arnold networks (KANs), constitutive KANs combine energy modeling, partial monotonicity, and symbolic interpretation [37], whereas monotonic input-convex KANs (ICKANs) add the convexity constraints used in their polyconvex invariant composition [38].

The features determine which responses this constrained core can represent. Let $\mathbf{z}$ be the feature vector and H a nondecreasing core. For two deformation states,

$$\mathbf{z}(\mathbf{F}_a) \leq \mathbf{z}(\mathbf{F}_b) \implies H(\mathbf{z}(\mathbf{F}_a)) \leq H(\mathbf{z}(\mathbf{F}_b)),$$

where the feature inequality is componentwise. A target core response with the opposite ordering cannot be fitted by increasing the network width. Additional reference and volumetric terms must be included before

applying this reasoning to the complete energy, but they do not remove the core’s ordering restriction. This simple observation makes feature design consequential before optimization begins.

Klein et al. [8] provide a pertinent example: their invariant model struggles with shear in a cubic lattice, while a deformation-gradient-based polyconvex model fits that response more accurately. The invariant approach performs well on their transversely isotropic example. The contrast identifies a representation issue, rather than a general failure of polyconvexity. More recent generalized-structure-tensor models learn aspects of anisotropy [39], and objective triclinic polyconvex formulations can be constructed without prescribing a nontrivial material symmetry group [40]. Accordingly, the question addressed here is not whether anisotropy can be learned at all, but how a particular admissible feature family can gain flexibility while retaining a compatible stress-free reference.

Table 1 positions this question relative to the constitutive and reduced-model formulations used as antecedents. It separates the object being approximated from the structure imposed on that object. This matters for the comparison: reducing microscopic equilibrium, differentiating a learned energy, and imposing polyconvexity are different operations, with different consequences. The table records those distinctions at formulation level.

Table 1: Positioning of the present study relative to neural constitutive models and projection-based reduced-order models. Properties refer to the specified formulation and its hypotheses, not to all models sharing an architecture name.

Reference / formulation	Represented object	Imposed structure	Scope relevant to this work
As’ad et al. [7]	Scalar energy in objective Green strain; stress from differentiation	Potential structure; ICNN convexity in Green strain	Stability depends on stress state; strain convexity is not a global polyconvexity certificate.
Xu et al. [11]	Incremental stress update using a learned lower-triangular matrix factor	The product $\mathbf{L}\mathbf{L}^T$ is positive semidefinite, and positive definite if $\mathbf{L}$ is nonsingular	Not necessarily an energy Hessian; the complete update derivative includes state-dependence terms.
Klein et al. [8]: invariant model	Convex nondecreasing energy network acting on polyconvex invariants	Objectivity and polyconvexity by construction for the selected invariants	Feature choice restricts approximation; the lattice example illustrates why enlarging the core can be insufficient.
Klein et al. [8]: gradient model	Network of deformation minors with material-group symmetrization	Polyconvexity and exact selected finite-group symmetry	Objectivity is approximate. The transverse-isotropy example also approximates the continuous symmetry by a finite subgroup.
Linden et al. [9]	Invariant energy network with analytic growth and reference corrections	Compatible polyconvex normalization for the stated isotropic and transverse-isotropic cases	Global nonnegative energy is separate from zero reference energy and stress.
Linka and Kuhl [35]	CANN using invariant energy terms and interpretable learned coefficients	Architecture encodes constitutive restrictions for the specified building blocks	Model discovery is within a specified invariant family, not a completeness result for arbitrary anisotropy.
Tac et al. [36]	Scalar NODE flows construct monotone invariant-energy derivatives	Integrable invariant expansion and polyconvexity under the stated nonnegative/monotone construction	Tissue examples are incompressible with selected fiber invariants; compressible extensions are discussed.
Ghaderi et al. [5]	Directional learning agents, microsphere summation, and maximum-history variables	Physics-based decomposition for polymer elasticity and inelastic effects	Includes history variables. Weight signs alone do not establish a general convexity guarantee.
Abdolazizi et al. [37]	CKAN scalar energy; spline edges, reference correction, and symbolic simplification	Energy differentiation and partial monotonicity in selected inputs	Partial monotonicity does not establish input convexity. Polyconvexity is identified as a possible extension.
Kalina et al. [39]	Invariants formed with learned generalized structure tensors of orders two, four, or six	Objective energy representation and identification of anisotropy through the tensor construction	Learns anisotropy; polyconvexity requires additional conditions on the invariants.
Klein et al. [40]	Objective triclinic polyconvex features, with optional finite-group symmetrization	Polyconvexity and objectivity without requiring a prescribed nontrivial material symmetry	Existing group-unspecified construction; the present focus is paired features with an isotropic reference derivative.
Thakolkaran et al. [38]	Input-convex monotone spline KAN for isotropic compressible energy	Polyconvex composition on the constrained spline domains	Finite-secant tails require a separate join check; our integrated-hat realization uses analytic C^2 matching.

Table 1 continued

Reference / formulation	Represented object	Imposed structure	Scope relevant to this work
Barnett et al. [21]; Ares De Parga et al. [22]	Primary/secondary state partition; secondary coordinates learned from primary coordinates	Low-dimensional closure integrated into a projection-based equilibrium or evolution model	Retains a reduced solve. Direct strain substitution is a separate operating mode here.
Farhat et al. [25]	Fixed-weight ECSW applied to reduced nonlinear finite element dynamics	Lagrangian structure preservation under the specified hypotheses	Does not automatically cover state-dependent weights or independent macroscopic stress cubature.
Hernández et al. [31]	Nonlinear-manifold virtual work with manifold-adaptive empirical cubature	Adaptive integration constraints and reduced equilibrium	Supplies the adaptive cubature used here. Integration accuracy and macroscopic constitutive guarantees remain distinct.

1.4 Present contribution

The constitutive contribution is a paired directional feature family with trainable orientations, stretch powers, and determinant exponents subject to explicit joint-convexity bounds. Pairing is chosen so that each feature’s strain derivative at the undeformed state is proportional to the identity tensor. The resulting reference stress can then be cancelled by a single scalar correction affine in the determinant, while the positive volumetric stiffness terms remain independently adjustable. Away from the reference state, the paired terms can still produce an anisotropic response. The construction thus links representational flexibility to a normalization that preserves polyconvexity.

The same feature family is coupled to a smooth ICNN and an integrated-hat spline ICKAN. Both realizations enforce objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and the specified growth behavior. For the spline realization, the tails have analytically matched derivatives. Nonnegative energy is treated separately through a sufficient lower bound evaluated for the selected trained parameters. The construction builds on existing ICNN and ICKAN principles and affine-determinant normalization; the specific contribution is their realization with this paired anisotropic family and the accompanying analytical conditions.

The computational contribution is a common assessment of PANNs and PROM-based surrogates against nested full-order FE^2 (FOM- FE^2). The affine HPROM, iterative HPROM-ANN, and direct decoder mode quantify the consequences of different levels of online microscopic approximation. POD, latent-space closure, ECM, and manifold-adaptive cubature are established ingredients, not new algorithms introduced here. Their inclusion makes it possible to assess whether the additional cost of retaining microscopic equilibrium yields useful structural accuracy compared with direct constitutive evaluation. Figure 1 summarizes the resulting alternatives.

1.5 Assessment strategy and scope

The numerical examples are organized around two questions. The periodic-cell study asks whether the constrained feature family can represent the homogenized anisotropic response and what the imposed constraints add beyond pointwise stress accuracy. Four constitutive tiers are considered: direct stress Regression, an unrestricted scalar-energy model termed Free, and the ICNN and ICKAN energies. Closed-cycle work tests integrability, while rank-one curvature addresses a condition not implied by potential structure alone. This attention to constitutively relevant derivatives is also motivated by the benchmarking study of Tac et al. [41]. The tensile-coupon study then measures how these constitutive approximations and the reduced microscopic models propagate into structural fields and complete online wall time. Field comparisons complement aggregate errors [42].

The assessment concerns one periodic material under plane-strain kinematics, a prescribed training domain, and one structural loading path. The coupon is a deployment of the same RVE, not a second material validation. Within this scope, the study demonstrates compatibility between accurate fitting and the proposed polyconvex construction; it does not establish universality across microstructures, three-dimensional constitutive completeness, or history-dependent behavior. The effects of feature learning, reference normalization, and network parameterization are assessed jointly in the proposed construction. Attributing the gain separately to feature learning, determinant exponents, normalization, or spline curvature requires matched-budget ablations.

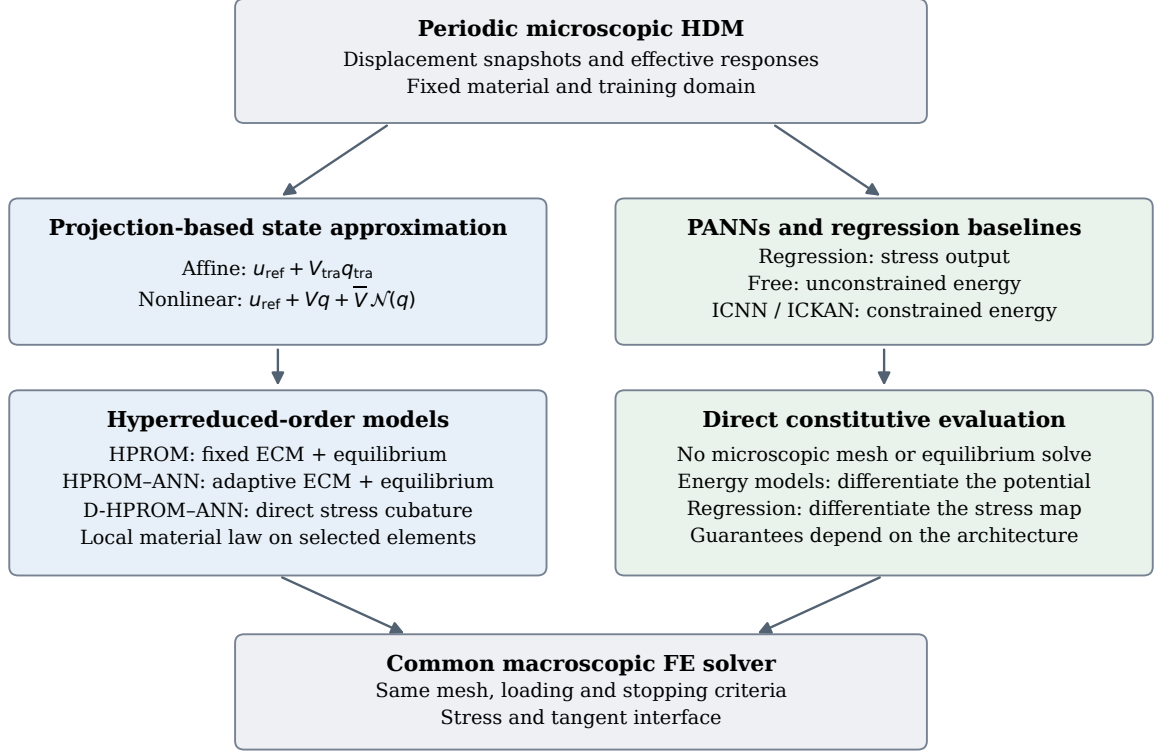


Figure 1: PROM-based and neural constitutive surrogates within a common macroscopic solver. PROM-based models reconstruct microscopic displacements and evaluate the constituent law; energy-based PANNs directly represent the homogenized potential. Regression and Free provide comparison baselines. The diagram distinguishes reduced equilibrium solves from direct evaluation.

Constitutive counterexamples are interpreted within their reference coverage. A nonzero closed-cycle work exposes nonconservativity in Regression, whereas sampled curvature tests complement, but do not replace, an analytical guarantee. The present extrapolative Free-model curvature witness lacks a converged FOM reference and is reported as a model-level observation, not as an FOM-validated spurious instability. Successful coupon convergence likewise does not certify a constitutive property. Finally, the online cost comparison excludes offline training and data-generation costs and reports the actual process/thread configurations and repetition counts. It compares the implemented online workflows under the reported configurations; it does not constitute a matched-resource statistical ranking or an offline-inclusive break-even analysis. Memory savings are a motivation for model reduction, but are not quantified in this assessment.

Section 2 defines the homogenization problem and work-conjugate variables. Section 3 develops the PROM-based surrogates, and Section 4 presents the learned constitutive laws. Sections 5 and 6 report the constitutive and structural assessments, respectively; Section 7 closes the paper.

2 Finite-strain homogenization and the reference problem

2.1 Kinematics and work-conjugate variables

We consider an in-plane deformation gradient $\mathbf{F} \in \text{GL}^+(2)$, with

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad J = \det \mathbf{F} > 0, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}). \quad (1)$$

Plane strain means that the three-dimensional gradient is $\text{diag}(\mathbf{F}, 1)$; it does not mean that the out-of-plane stress vanishes. The engineering strain vector and its work-conjugate stress vector are

$$\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (S_{11}, S_{22}, S_{12})^T, \quad \mathbf{S} : d\mathbf{E} = \mathbf{s} \cdot d\mathbf{e}. \quad (2)$$

We also write $\gamma_{12} = 2E_{12}$. For an objective differentiable energy W , the second Piola stress and constitutive tangent are $\mathbf{s} = \partial_{\mathbf{e}} W$ and $\mathbf{D} = \partial_{\mathbf{e}\mathbf{e}}^2 W$. The factors in Eq. (2) are essential for both energy-cycle tests and derivative checks.

2.2 Periodic microscopic equilibrium

Let Y denote a reference periodic cell and $Y_s \subset Y$ its solid part. A homogeneous macroscopic gradient $\bar{\mathbf{F}}$ induces the microscopic deformation

$$\boldsymbol{\varphi}(\mathbf{X}) = \bar{\mathbf{F}}\mathbf{X} + \mathbf{w}(\mathbf{X}), \quad \mathbf{F}_\mu = \bar{\mathbf{F}} + \nabla \mathbf{w}, \quad (3)$$

where $\mathbf{w}$ is periodic on opposite faces. A constraint removes its translational null space, and the void surface is traction-free. On a differentiable equilibrium branch, the discrete fluctuation coordinates $\mathbf{a}^*(\bar{\mathbf{F}})$ satisfy

$$\mathbf{r}(\mathbf{a}, \bar{\mathbf{F}}) = \partial_{\mathbf{a}} \Pi_\mu(\mathbf{a}, \bar{\mathbf{F}}) = \mathbf{0}, \quad \Pi_\mu = \int_{Y_s} \psi_\mu(\mathbf{F}_\mu) dV. \quad (4)$$

The branch energy and homogenized first Piola stress are

$$\bar{W} = \frac{\Pi_\mu(\mathbf{a}^*, \bar{\mathbf{F}})}{|Y|}, \quad \bar{\mathbf{P}} = \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu dV, \quad \bar{\mathbf{S}} = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}. \quad (5)$$

The denominator includes the void. Averaging microscopic second Piola stress instead of first Piola stress would generally give a different quantity and is not used. Stationarity in Eq. (4) eliminates the fluctuation derivative in the first derivative of $\bar{W}$. More explicitly, periodic test fluctuations satisfy

$$\int_{Y_s} \mathbf{P}_\mu : \nabla \delta \mathbf{w} dV = 0, \quad \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu : \delta \mathbf{F}_\mu dV = \bar{\mathbf{P}} : \delta \bar{\mathbf{F}}. \quad (6)$$

The second identity follows from the first and the additive microkinematics. It is the work-consistency condition of the periodic micro-to-macro transition [1]. Void boundaries contribute no traction work. The cell is an ideal periodic material model; no claim of statistical representativeness for a random microstructure is needed.

The implementation queries the constitutive law through $\bar{\mathbf{F}} = \sqrt{\mathbf{I} + 2\bar{\mathbf{E}}}$, exploiting objectivity to remove a superposed macroscopic rotation. The microscopic Newton method follows an equilibrium branch; convergence does not establish global minimization of Π_μ . In particular, no general assertion that homogenization preserves polyconvexity is required here.

2.3 Consistent tangent and the scope of the reference

If $\mathbf{r}_{,\mathbf{a}}$ is nonsingular on the selected branch, implicit differentiation gives

$$\frac{d\mathbf{a}^*}{d\mathbf{e}} = -\mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}, \quad \mathbf{D} = \mathbf{s}_{,\mathbf{e}} - \mathbf{s}_{,\mathbf{a}} \mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}. \quad (7)$$

Every derivative in this expression must refer to the stress and residual actually evaluated. The reference and reduced implementations are checked against perturbations that re-solve microscopic equilibrium. These checks validate linearization, not a change of homogenization assumptions.

The nested FOM-FE² solution is the reference for surrogate errors at fixed macro and micro discretizations. It is not a direct numerical simulation of all pores in the coupon. Homogenization error, boundary-layer effects, and mesh-discretization error therefore remain conceptually distinct from the surrogate errors reported below.

2.4 Macroscopic equilibrium and the constitutive interface

Let $\mathbf{U}$ collect the free macroscopic nodal displacements, and let $\mathbf{e}_g(\mathbf{U})$ be the engineering Green strain at macroscopic integration point g . With $\mathbf{B}_g = \partial \mathbf{e}_g / \partial \mathbf{U}$, reference integration weights Ω_g , and dead external loads $\mathbf{f}_{\text{ext}}$, equilibrium reads

$$\mathbf{R}_M(\mathbf{U}) = \sum_g \Omega_g \mathbf{B}_g^T \mathbf{s}_g(\mathbf{e}_g(\mathbf{U})) - \mathbf{f}_{\text{ext}} = \mathbf{0}. \quad (8)$$

Its tangent contains material and geometric contributions,

$$\mathbf{K}_M = \sum_g \Omega_g \left[\mathbf{B}_g^T \mathbf{D}_g \mathbf{B}_g + \sum_{\alpha=1}^3 (s_g)_\alpha \frac{\partial^2 (e_g)_\alpha}{\partial \mathbf{U} \partial \mathbf{U}} \right]. \quad (9)$$

The first term uses the derivative of the deployed stress law; the second follows from nonlinear strain kinematics. The constitutive interface therefore requires both $\mathbf{s}_g$ and $\mathbf{D}_g$. It does not require the internal implementation to use a microscopic mesh, reduced coordinates, or a neural energy.

In FOM-FE², this interface invokes the full periodic solve. In the HPROM variants it invokes a reduced microscopic evaluator. In the learned constitutive variants it evaluates the stress and tangent map directly. The macroscopic mesh, loading and equilibrium algorithm are kept unchanged. The suffix FE² used in the archived model names identifies this common multiscale comparison; direct neural evaluations are not nested microscopic finite element solves.

3 Projection-based reduced-order models

3.1 High-dimensional state and periodic lifting

We use the PMOR nomenclature of Ares De Parga et al. [22]: the high-dimensional model (HDM) has state dimension N ; a PROM solves for generalized coordinates in a reduced approximation space; an HPROM additionally approximates the evaluation of its projected operators. The acronym FOM is retained in FOM-FE² to identify the archived full-order reference. HDM and FOM refer to the same unreduced microscopic discretization here.

It is useful to distinguish the physical nodal displacement vector $\mathbf{d} \in \mathbb{R}^{N_{\text{dof}}}$ from the independent state $\mathbf{u} \in \mathbb{R}^N$ obtained after enforcing periodicity and removing rigid translation. In the deployed convention,

$$\mathbf{d}(\mathbf{u}, \mathbf{e}) = \mathbf{T}\mathbf{u} + \mathbf{g}_{\text{per}}(\mathbf{e}), \quad \mathbf{R}(\mathbf{u}; \mathbf{e}) = \mathbf{T}^T \mathbf{f}_{\text{int}}(\mathbf{d}(\mathbf{u}, \mathbf{e})) = \mathbf{0}. \quad (10)$$

Here $\mathbf{T}$ is the fixed periodic identification operator and $\mathbf{g}_{\text{per}}$ supplies the strain-dependent displacement jumps and pinned values. The residual is written with the internal-force sign convention; reversing the residual sign leaves the equilibrium and implicit sensitivities unchanged. The HDM tangent is $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R}$.

The snapshots retain the affine part of the independent nodal displacement. They are therefore not snapshots of $\mathbf{a}$ in Eq. (4). If $\mathbf{u}_{\text{aff}}(\mathbf{e})$ denotes the affine displacement at the independent nodes, then $\mathbf{a} = \mathbf{u} - \mathbf{u}_{\text{aff}}(\mathbf{e})$ relates the two conventions. Both generate the same physical field when the lifting is transformed consistently. Using a basis trained in one convention with the lifting of the other would define a different and generally incorrect reduced problem. Below, $\mathbf{u}$ always denotes the independent, affine-retaining state of Eq. (10).

3.2 POD approximation and the conventional PROM

Let $\mathbf{u}^s$ be the HDM solution at a sampled macroscopic strain $\mathbf{e}^s$, with $s = 1, \dots, N_s$. Given a reference state $\mathbf{u}_{\text{ref}}$, form

$$\mathbf{S}_{\text{snap}} = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}}, \dots, \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}] = \mathbf{U}_S \mathbf{\Sigma}_S \mathbf{Y}_S^T. \quad (11)$$

The subscript distinguishes this matrix from the second Piola stress $\mathbf{S}$. The columns of $\mathbf{U}_S$ associated with the largest singular values generate a POD reduced-order basis (ROB). For an orthonormal n_{tra} -column basis $\mathbf{V}_{\text{tra}}$, the conventional affine approximation is

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}, \quad \mathbf{q}_{\text{tra}} \in \mathbb{R}^{n_{\text{tra}}}, \quad \mathbf{V}_{\text{tra}}^T \mathbf{V}_{\text{tra}} = \mathbf{I}. \quad (12)$$

Here *linear PROM* refers to this affine approximation space, not to a linear constitutive law or a linear equilibrium system.

For a Euclidean POD, the relative squared snapshot reconstruction error is

$$\epsilon_{\text{POD}}^2(n_{\text{tra}}) = \frac{\sum_{j > n_{\text{tra}}} \sigma_j^2}{\sum_j \sigma_j^2}. \quad (13)$$

This is a compression criterion for the supplied snapshots, not an error bound for stresses at unsampled strains or for the eventual coupon solution. Force evaluation, constitutive differentiation, and equilibrium conditioning can make small displacement components important.

Galerkin projection gives

$$\mathbf{r}_{\text{tra}}(\mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{V}_{\text{tra}}^T \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{0}, \quad \mathbf{K}_{\text{tra}} = \mathbf{V}_{\text{tra}}^T \mathbf{K} \mathbf{V}_{\text{tra}}. \quad (14)$$

The system has n_{tra} unknowns, but assembling $\mathbf{R}$ and $\mathbf{K}$ on every microscopic element still scales with the full discretization. This is the cost removed by hyperreduction, not by POD alone.

3.3 Primary and secondary spaces: the PROM-ANN

Following the latent-space closure formulation [21, 22], introduce mutually orthogonal primary and secondary ROB, s,

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad \bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{n}}, \quad (15)$$

$$\mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}, \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad \bar{\mathbf{V}}^T \bar{\mathbf{V}} = \mathbf{I}. \quad (16)$$

The associated primary and secondary generalized coordinates are

$$\mathbf{q}^s = \mathbf{V}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{q}}^s = \bar{\mathbf{V}}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}). \quad (17)$$

Instead of solving online for both vectors, a closure map $\mathcal{N} : \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ is learned from pairs $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$. The approximation becomes

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}; \boldsymbol{\eta}), \quad \mathbf{B}(\mathbf{q}) = \partial_{\mathbf{q}} \tilde{\mathbf{u}} = \mathbf{V} + \bar{\mathbf{V}}\mathcal{N}_{,\mathbf{q}}. \quad (18)$$

Only the n primary coordinates are online equilibrium unknowns. The $\bar{n}$ secondary coordinates remain in the displacement reconstruction; they are not independent unknowns and are not discarded physically. In the terminology of the source framework, the closure recovers information in modes discarded from the primary approximation.

For orthonormal blocks, the error at a snapshot separates into closure error and a component outside the total ROB:

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(\mathbf{I} - \mathbf{V}_{\text{tot}}\mathbf{V}_{\text{tot}}^T)(\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (19)$$

This explains the low-dimensional targets: the ANN approximates secondary coordinates, while the ROB performs the displacement reconstruction. It also separates limitations that cannot be repaired by the same operation. Increasing network capacity cannot recover a component excluded from $\mathbf{V}_{\text{tot}}$; adding secondary modes cannot make a non-single-valued primary-to-secondary relation single-valued.

The deployed closure minimizes the unscaled secondary-coordinate mean-square error,

$$\mathcal{L}_{\text{cl}}(\boldsymbol{\eta}) = \frac{1}{|\mathcal{I}_{\text{fit}}|\bar{n}} \sum_{s \in \mathcal{I}_{\text{fit}}} \|\mathcal{N}(\mathbf{q}^s; \boldsymbol{\eta}) - \bar{\mathbf{q}}^s\|_2^2. \quad (20)$$

Input standardization is absorbed into the closure map, but the secondary outputs are not standardized mode by mode. Because $\bar{\mathbf{V}}$ is orthonormal, this loss measures the displacement error within the secondary space, up to its constant normalization. Equalizing relative errors of individual modes would optimize a different metric. Neither objective alone is a stress-error guarantee; the reconstructed state must also be assessed through the microscopic constitutive output.

The total ROB need not have the same dimension as a separately selected conventional PROM. Here both models use the same retained 39-dimensional span, with $n = 3$ and $\bar{n} = 36$ for the nonlinear model. The latter reduces equilibrium unknowns while retaining the available displacement span, but represents a nonlinear subset of that span rather than all of it.

3.4 The strain-informed change of primary coordinates

The deployed primary directions are not simply the first three POD modes. They are obtained by rotating the retained span so that its primary coefficients are informative about the three macroscopic strain components. Let $\boldsymbol{\Phi}$ be the initial orthonormal POD basis, and define

$$\mathbf{Q}_{\text{POD}} = \boldsymbol{\Phi}^T \mathbf{S}_{\text{snap}}, \quad \mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}], \quad \mathbf{T}_m = \mathbf{M} \mathbf{Q}_{\text{POD}}^\dagger. \quad (21)$$

The superscript $\dagger$ denotes the Moore–Penrose pseudoinverse. A compact SVD,

$$\boldsymbol{\Phi} \mathbf{T}_m^T = \mathbf{V} \boldsymbol{\Sigma}_m \mathbf{Z}_m^T, \quad \mathbf{A}_m = \boldsymbol{\Sigma}_m^{-1} \mathbf{Z}_m^T, \quad (22)$$

defines the primary orthonormal block, assuming rank three. The secondary block is an energy-ordered orthonormal complement in the retained POD span. The implementation uses coordinates $\boldsymbol{\xi} = \mathbf{A}_m^{-1} \mathbf{q}$ and a closure $\hat{\mathcal{N}}(\boldsymbol{\xi})$, so Eq. (18) is evaluated as

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \boldsymbol{\xi} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \mathcal{N}(\mathbf{q}) = \hat{\mathcal{N}}(\mathbf{A}_m^{-1} \mathbf{q}). \quad (23)$$

For projected training states, $\boldsymbol{\xi}^s = \mathbf{T}_m \mathbf{Q}_{\text{POD}}^s$ approximates $\mathbf{e}^s$. This is a fitted relation, not an identity of continuum kinematics. The iterative HPROM-ANN solves for $\boldsymbol{\xi}$ and need not return $\boldsymbol{\xi} = \mathbf{e}$. The direct variant below makes that additional substitution explicitly. This preserves the nomenclature of Eq. (18) without hiding the implemented change of coordinates.

3.5 Projection on the tangent space and linearization

For nonlinear Galerkin projection, virtual displacements belong to the tangent space of the manifold. The reduced equilibrium is

$$\mathbf{r}(\mathbf{q}; \mathbf{e}) = \mathbf{B}(\mathbf{q})^T \mathbf{R}(\tilde{\mathbf{u}}(\mathbf{q}); \mathbf{e}) = \mathbf{0}. \quad (24)$$

Its exact derivative is

$$\frac{\partial \mathbf{r}}{\partial \mathbf{q}} = \mathbf{B}^T \mathbf{K} \mathbf{B} + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \mathbf{q} \partial \mathbf{q}}. \quad (25)$$

The second term accounts for manifold curvature. It need not vanish at reduced equilibrium, because $\mathbf{B}^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$. Dropping it defines an approximate iteration matrix, not the exact Jacobian.

The closure construction does not prescribe the projection method. Least-squares Petrov–Galerkin (LSPG) minimizes a discrete residual norm and uses a different test space, involving the residual Jacobian and decoder tangent [19, 22, 42]. The periodic hyperelastic problem here uses Galerkin virtual work. Thus the notation and latent closure are inherited from PROM–ANN, but the LSPG equations of its CFD applications are not copied into this mechanics problem.

3.6 Fixed element-selected ECM: the HPROM

Let $\mathcal{E}$ be the microscopic element set and $\mathbf{L}_e$ extract element nodal values from $\mathbf{d}$. Define

$$\mathbf{d}_e = \mathbf{L}_e(\mathbf{T}\tilde{\mathbf{u}} + \mathbf{g}_{\text{per}}), \quad \mathbf{B}_e = \mathbf{L}_e \mathbf{T} \mathbf{B}. \quad (26)$$

The reduced force is a sum of contributions $\mathbf{h}_e = \mathbf{B}_e^T \mathbf{f}_{\text{int},e}$. Fixed ECM gives

$$\mathbf{r}_{\text{ECM}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e \mathbf{h}_e(\mathbf{q}; \mathbf{e}) = \mathbf{0}, \quad \mathcal{Z}_r \subset \mathcal{E}, \quad w_e \geq 0. \quad (27)$$

For the conventional HPROM, $\mathbf{B}$ is replaced by $\mathbf{V}_{\text{tra}}$ and the unknown is $\mathbf{q}_{\text{tra}}$. There is no ANN in this tier. A selected element retains its ordinary within-element quadrature. Consequently, selected elements, selected Gauss points, and independent coordinates are distinct counts.

Empirical cubature seeks a sparse nonnegative rule reproducing a collection of reduced integrands. Writing their sampled values in a matrix $\mathbf{G}_{\text{cub}}$, with one column per candidate element, the fixed-rule objective is schematically

$$\min_{\mathbf{w} \geq 0} \|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2, \quad \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}, \quad (28)$$

with a support-selection strategy and integration tolerance. ECM uses a compressed representation of the integrands [26]. Its training concerns individual contributions, not merely the vanishing assembled equilibrium residual: training only on assembled zero residuals would contain no useful cubature information. Constant-integration conditions can be included separately. The resulting support and weights are fixed online.

3.7 Manifold-adaptive weights: the HPROM–ANN

For nonlinear kinematics, both the decoded state and its tangent enter the element integrands. The adaptive-weight rule [31] is

$$\mathbf{r}_{\text{MAW}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e(\mathbf{q}) \mathbf{B}_e(\mathbf{q})^T \mathbf{f}_{\text{int},e}(\mathbf{d}_e) = \mathbf{0}. \quad (29)$$

The support is fixed; the weights vary with the primary state. The deployed parametrization is

$$w_e(\mathbf{q}) = N_e \frac{\exp(\ell_e(\mathbf{q}))}{\sum_{j \in \mathcal{Z}_r} \exp(\ell_j(\mathbf{q}))}, \quad \sum_{e \in \mathcal{Z}_r} w_e = N_e, \quad (30)$$

where $N_e = |\mathcal{E}|$ and ℓ_e are the learned logits, evaluated in deployed coordinates when appropriate. N_e normalizes element multipliers; physical quadrature measures remain in the element contributions. Positivity and the prescribed sum hold at a query, but integration accuracy remains an approximation property.

For a selected support, let $\mathbf{A}^s$ contain the selected element contributions at training state s , and let $\mathbf{b}^s$ be the corresponding full-mesh sum. The weight-network training objective is

$$\mathcal{L}_{\text{cub}} = \frac{1}{|\mathcal{I}_{\text{fit}}|} \sum_{s \in \mathcal{I}_{\text{fit}}} \frac{\|\mathbf{A}^s \mathbf{w}(\mathbf{q}^s) - \mathbf{b}^s\|_2^2}{\max(\|\mathbf{b}^s\|_2^2, \delta_{\text{floor}})}. \quad (31)$$

The small positive floor avoids division by zero. A preliminary fit to per-state constrained weights initializes the network; the final objective concerns integration error, not agreement with a unique target weight vector. Different weight vectors can integrate the same target. Residual and stress fields are trained separately, and the mean-square training loss must be distinguished from a median relative integration error used for checkpoint selection or reporting.

Differentiation gives

$$\mathbf{r}_{\text{MAW},\mathbf{q}} = \sum_{e \in \mathcal{Z}_r} \left[w_e \mathbf{B}_e^T \mathbf{K}_e \mathbf{B}_e + w_e \sum_j (f_{\text{int},e})_j \partial_{\mathbf{q}\mathbf{q}}^2 (d_e)_j + \mathbf{h}_e (\nabla_{\mathbf{q}} w_e)^T \right]. \quad (32)$$

Besides projected stiffness, the exact derivative contains decoder curvature and weight variation. The microscopic iteration uses the first term as a modified-Newton matrix and refreshes the decoder and weights at each state. The macroscopic constitutive tangent instead uses derivatives of the complete residual and output, including the additional terms. Central differences evaluate these derivatives without explicitly assembling a third-order decoder tensor.

3.8 Stress reconstruction and the direct variant

The output uses a separate cubature support $\mathcal{Z}_s$ and weights v_e . With microscopic first Piola stress $\mathbf{P}_{\mu,g}$ and reference quadrature measure ω_g ,

$$\bar{\mathbf{P}}_h(\mathbf{q}, \mathbf{e}) = \frac{1}{|Y|} \sum_{e \in \mathcal{Z}_s} v_e(\mathbf{q}) \sum_{g \in e} \omega_g \mathbf{P}_{\mu,g}(\mathbf{d}_e), \quad \bar{\mathbf{S}}_h = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}_h. \quad (33)$$

For HPROM, v_e is constant; for HPROM-ANN it is adaptive. Residual and stress supports can overlap, and their union specifies the element data needed online. Independently training the output rule allows its accuracy to be controlled separately from equilibrium.

For a converged reduced state $\mathbf{q}^*(\mathbf{e})$, the deployed constitutive tangent is

$$\mathbf{D}_h = \mathbf{s}_{h,e} - \mathbf{s}_{h,\mathbf{q}} \mathbf{r}_{h,\mathbf{q}}^{-1} \mathbf{r}_{h,e}. \quad (34)$$

Partial derivatives hold the other argument fixed. In particular, $\mathbf{s}_{h,e}$ includes the periodic lifting and the first-to-second-Piola conversion. Central differences of the actual residual and output are combined through the small linear system in Eq. (34). This avoids an additional nonlinear microscopic solve for each macroscopic strain perturbation.

D-HPROM-ANN removes reduced equilibrium by prescribing $\boldsymbol{\xi} = \mathbf{e}$ in Eq. (23):

$$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathbf{N}}(\mathbf{e}). \quad (35)$$

This state is inserted directly into the adaptive stress rule. Its tangent differentiates the complete direct map, including weights and lifting. The prefix D identifies this operating mode; it is not a projection method from Ref. [22]. With no reduced equilibrium solve, accuracy depends directly on the learned relation between loading and microscopic state. There is no general monotone error ordering between iterative and direct variants.

3.9 What mechanical structure does hyperreduction retain?

If microscopic forces derive from element energies, fixed-weight Galerkin projection with consistent kinematics can be derived from a weighted reduced energy. This underlies structure-preservation results for ECSW under their stated hypotheses [25]. Two features of the present deployment need separate attention.

First, independent equilibrium and stress cubatures need not define derivatives of one energy. Second, weighting an energy derivative differs from differentiating an energy with state-dependent weights:

$$\nabla_{\mathbf{q}} \sum_e w_e(\mathbf{q}) \Pi_e(\mathbf{q}, \mathbf{e}) = \sum_e w_e(\mathbf{q}) \nabla_{\mathbf{q}} \Pi_e + \sum_e \Pi_e \nabla_{\mathbf{q}} w_e. \quad (36)$$

The last term is absent from Eq. (29). Positive weights do not remove it. Accordingly, these models retain microscopic constitutive evaluations and, for iterative variants, projected equilibrium, but are not assigned an exact macroscopic potential or polyconvexity certificate without further analysis. A consistent tangent establishes differentiation of the implemented map; it does not make that map conservative.

3.10 Offline–online separation and reduced meshes

POD, closure fitting, cubature selection, and reduced-mesh construction are offline operations. Online, only the rows of the lifting and basis operators needed by the active supports are used. Element forces and stiffnesses are projected locally. Periodic lifting is evaluated only at selected nodes, and residual/output perturbations are batched where possible. D-HPROM–ANN omits decoder derivatives unnecessary for stress-only evaluations.

These operations are essential to the intended complexity reduction. Masking a full microscopic reconstruction or full-mesh assembly would leave a cost proportional to discarded elements. The deployed MDPA meshes contain the selected elements and required nodes; discarded elements are not assembled. Per-query cost therefore contains selected-element constitutive work, restricted decoding and, for iterative models, a reduced solve. It depends on support size, secondary dimension, iteration count, and batching, not only primary dimension.

Table 2: Objects retained online by the microscopic approximations. Stress is obtained from the microscopic law and first Piola averaging.

Model	State representation	Equilibrium unknowns	Integration	Learned object
HDM / FOM	$\mathbf{u} \in \mathbb{R}^N$	N	Full mesh	None
PROM	$\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$	n_{tra}	Full mesh	None
HPROM	Same affine space	n_{tra}	Fixed ECM	None
HPROM–ANN	Eq. (18)	n	Adaptive ECM	Secondary coordinates and weights
D-HPROM–ANN	Eq. (35)	None	Adaptive stress rule	Same closure, evaluated directly

4 Learned constitutive laws

4.1 Constitutive requirements and the variables they constrain

A constitutive approximation must be assessed at several levels. Objectivity concerns a change of observer; potential structure concerns work along a path; convexity conditions concern particular variations of deformation; reference normalization fixes the undeformed state. These requirements should not be collapsed into a single label such as physics-informed.

For a superposed proper rotation $\mathbf{Q}$, objectivity requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}). \quad (37)$$

Using $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as an input to an energy enforces the first identity. A direct map from $\mathbf{E}$ to the material stress $\mathbf{S}$ can also be objective without possessing a potential. Objectivity therefore does not distinguish Regression from Free in this study. Material symmetry is different: a right action $\mathbf{F} \mapsto \mathbf{F}\mathbf{Q}_m$ represents a symmetry of the reference material only for transformations in its material symmetry group. Directional features transform with the reference material, not with the observer.

For a differentiable hyperelastic energy, stress is its derivative in work-conjugate variables. In the convention of Eq. (2),

$$\mathbf{s}(\mathbf{e}) = \nabla_{\mathbf{e}} W(\mathbf{e}), \quad \oint_{\Gamma} \mathbf{s} \cdot d\mathbf{e} = 0. \quad (38)$$

The closed-path identity is exact for a single-valued differentiable potential. On a simply connected strain domain, a continuously differentiable stress map has such a potential if its Jacobian is symmetric. Direct stress regression need not impose the cross-derivative identities $\partial s_i / \partial e_j = \partial s_j / \partial e_i$ [7]. A small pointwise stress error alone does not establish them.

Potential structure does not impose a stability condition. The Legendre–Hadamard condition for a twice differentiable energy is

$$\partial_{\mathbf{F}\mathbf{F}}^2 W(\mathbf{F})[\mathbf{a} \otimes \mathbf{b}, \mathbf{a} \otimes \mathbf{b}] \geq 0 \quad \text{for all } \mathbf{a}, \mathbf{b}. \quad (39)$$

Strict positivity for nonzero directions is a stronger statement than this nonnegative condition. In two dimensions, a sufficient condition is a representation

$$W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F}), \quad \mathcal{P} \text{ convex in its independent arguments.} \quad (40)$$

This is two-dimensional polyconvexity. The cofactor is linear in the entries of a 2×2 matrix, so it need not be introduced as an additional independent argument. Along a rank-one line $\mathbf{F} + t\mathbf{a} \otimes \mathbf{b}$, both $\mathbf{F}$ and its determinant depend affinely on t . Convexity of $\mathcal{P}$ then implies convexity of the restriction of W to every admissible segment of that line.

The corresponding three-dimensional definition uses $(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F})$ [12, 13]. The restriction of a three-dimensional polyconvex law to plane-strain kinematics is compatible with a two-dimensional polyconvex representation. The converse does not follow: establishing Eq. (40) does not establish an admissible three-dimensional extension with arbitrary out-of-plane deformation. The proposed construction and its guarantees are stated here for the actual in-plane model.

Convexity in $\mathbf{E}$ is a different property. To see why, take an arbitrary perturbation $\mathbf{H}$ of $\mathbf{F}$. Along $\mathbf{F}(t) = \mathbf{F} + t\mathbf{H}$,

$$\dot{\mathbf{E}} = \text{sym}(\mathbf{F}^T \mathbf{H}), \quad \ddot{\mathbf{E}} = \mathbf{H}^T \mathbf{H}, \quad \frac{d^2 W}{dt^2} = \dot{\mathbf{e}}^T \mathbf{D} \dot{\mathbf{e}} + \mathbf{S} : (\mathbf{H}^T \mathbf{H}). \quad (41)$$

Even if $\mathbf{D}$ is positive semidefinite, the stress term can be negative in compression. This is the distinction behind the stress-dependent arguments in As’ad et al. [7] and Gao et al. [10]. Convexity in $\mathbf{C}$ and in $\mathbf{E}$ are equivalent up to their affine change of variables, but neither should be relabeled as polyconvexity. Similarly, a positive-matrix factor in an incremental stress update [11] is not automatically a Hessian of a scalar energy: the required compatibility between derivatives still has to be established.

Reference normalization imposes

$$W(\mathbf{I}) = 0, \quad \mathbf{P}(\mathbf{I}) = \mathbf{0}. \quad (42)$$

The first condition fixes an arbitrary additive energy constant. The second is a substantive stress condition and depends on how the energy is corrected. Subtracting a linear function of $\mathbf{E}$ preserves a potential but need not preserve polyconvexity in $\mathbf{F}$. An affine function of an independent deformation minor, in contrast, preserves convexity of $\mathcal{P}$. The normalization below exploits precisely this distinction.

Finally, the inequalities $W(\mathbf{F}) \geq 0$, the barrier $W \rightarrow \infty$ as $J \rightarrow 0^+$, and the existence of minimizers are not synonyms. Ciarlet’s presentation of Ball’s existence argument [13], especially Sections 4.9 and 7.7, includes coercivity, admissible boundary data, and finite-energy competitors in addition to polyconvexity. Neither the existence theory nor a constitutive barrier guarantees convergence of an unconstrained Newton step. In this work, architectural statements, saved-parameter bounds, and numerical observations are therefore identified separately.

4.2 Why the feature map matters before the network

Consider a feature vector $\mathbf{z}(\mathbf{F})$ and a core H nondecreasing in each input. For any two deformation states,

$$\mathbf{z}(\mathbf{F}_B) \geq \mathbf{z}(\mathbf{F}_A) \text{ componentwise} \implies H(\mathbf{z}(\mathbf{F}_B)) \geq H(\mathbf{z}(\mathbf{F}_A)). \quad (43)$$

This implication holds regardless of the network’s width. If the complete constitutive energy is $W = H(\mathbf{z}) + W_{\text{corr}}$, the ordered quantity is $W - W_{\text{corr}}$, not W alone. Thus a decreasing reference energy along such a path indicates an obstruction only after the complete correction is accounted for. The diagnosis must examine the feature map and analytic terms together.

This observation explains why changing an admissible feature construction can be more effective than increasing the capacity of its convex core. The original 15-feature model and the paired 32-feature model are not different numbers of independent strains: both represent the same three-dimensional engineering-strain state. They are different embeddings of that state into variables on which the network is constrained to be monotone and convex. Additional features can enlarge the admissible response family, but their number alone does not prove completeness.

The paired construction below couples directional stretch to volume through admissible determinant exponents. A feature can therefore decrease along a path when its denominator grows sufficiently, even if a numerator increases. Moreover, its reference derivative is controlled analytically, allowing normalization and volumetric stiffness to be set independently. These are changes to the representation, not changes to the training strain box.

4.3 What is changed between the four learned models?

Table 3 compares the actual model families used in this study. It is deliberately separate from the literature comparison: a property of our implementation is not automatically attributed to an entire class of published methods.

Table 3: Learned constitutive models in the present comparison. “Not imposed” is not synonymous with “fails at every state”.

	Regression	Free	ICNN	ICKAN
Network input	$E_{11}, E_{22}, \gamma_{12}$	$C_{11} - 1, C_{22} - 1, C_{12}, J - 1$	32 paired features	32 paired features
Network output	Three stresses	Scalar energy	Scalar energy core	Scalar energy core
Potential structure	Not imposed	Exact	Exact	Exact
Objectivity	Objective strain input and material stress output	Exact	Exact	Exact
Reference energy/stress	Not imposed	Exact normalization	Exact normalization	Exact normalization
Polyconvexity	Not imposed	Not imposed	By construction	By construction
Global $W \geq 0$	No potential	Not guaranteed	Separate saved-weight bound	Separate saved-weight bound

Regression directly approximates $\mathbf{e} \mapsto \mathbf{s}$ with three outputs. The Free model instead defines an unconstrained energy network $\widetilde{W}(\mathbf{e})$ and applies

$$W_{\text{free}}(\mathbf{e}) = \widetilde{W}(\mathbf{e}) - \widetilde{W}(\mathbf{0}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0}) \cdot \mathbf{e}. \quad (44)$$

This enforces zero reference energy and stress without enforcing polyconvexity. The four Free inputs are redundant coordinates of the same three-dimensional strain state; they are not four independent loading parameters. Its use of material-coordinate components of $\mathbf{C}$ does not impose isotropy. Reference-axis changes must transform the material directions together with the tensor components.

4.4 Paired directional features

For each feature, let $\mathbf{d}_i = (\cos \theta_i, \sin \theta_i)^T$ and $\mathbf{d}_i^\perp = (-\sin \theta_i, \cos \theta_i)^T$. Define

$$z_i(\mathbf{F}, J) = \frac{\|\mathbf{F}\mathbf{d}_i\|^{2p_i}}{p_i J^{b_i}} + \frac{\|\mathbf{F}\mathbf{d}_i^\perp\|^{2q_i}}{q_i J^{c_i}}, \quad (45)$$

with

$$p_i, q_i \geq \frac{1}{2}, \quad 0 \leq b_i \leq 2p_i - 1, \quad 0 \leq c_i \leq 2q_i - 1. \quad (46)$$

The parameters $\theta_i, p_i, q_i, b_i, c_i$ are learned subject to Eq. (46). The powers use a shifted softplus parametrization, and the determinant exponents use sigmoid fractions of their admissible upper bounds. The paired terms need not have equal exponents. Consequently, pairing perpendicular directions does not impose invariance under their interchange.

Proposition 1. *Each feature in Eq. (45) is objective and convex in the independent variables $(\mathbf{F}, J)$ on $J > 0$. Its value and engineering-strain gradient at the identity are*

$$z_i^0 = \frac{1}{p_i} + \frac{1}{q_i}, \quad \partial_{\mathbf{e}} z_i|_{\mathbf{0}} = \rho_i(1, 1, 0)^T, \quad \rho_i = 2 - \frac{b_i}{p_i} - \frac{c_i}{q_i}. \quad (47)$$

Proof. The map $\|\mathbf{x}\|^a J^{-b}$ is jointly convex for $a \geq 1$ and $0 \leq b \leq a - 1$, as shown in Appendix A. Composition with the linear maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$ proves convexity. Norms are unchanged by left rotations, proving objectivity. At the identity, differentiating the two numerators yields $2\mathbf{d}_i \otimes \mathbf{d}_i$ and $2\mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp$ with respect to $\mathbf{E}$. Their sum is $2\mathbf{I}$. The determinant factors contribute $-(b_i/p_i + c_i/q_i)\mathbf{I}$, giving Eq. (47). $\square$

Positive feature scaling and subtraction of a reference feature vector improve numerical conditioning without changing these properties. They are constants with respect to deformation for every frozen parameter set. Learning the directions does not amount to identifying or enforcing a complete material symmetry group. No completeness or universal-approximation theorem is asserted for a finite collection of these features followed by a monotone convex core.

4.5 Reference normalization, growth, and polyconvexity

Let $H(\mathbf{z})$ denote a differentiable convex function that is nondecreasing in every feature, absorbing the positive scaling and centering into its definition. Set

$$h_i = \partial_{z_i} H(\mathbf{z}^0) \geq 0, \quad r = \sum_i h_i \rho_i. \quad (48)$$

The energy is

$$\begin{aligned} W(\mathbf{F}) = & H(\mathbf{z}) - H(\mathbf{z}^0) - r(J - 1) + \alpha(J - 1 - \log J) + \frac{\beta}{2}(J - 1)^2 \\ & + \varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right), \quad \alpha, \beta, \varepsilon > 0. \end{aligned} \quad (49)$$

Affine determinant stress normalization has an established precedent in Linden et al. [9]. The role of Eq. (47) here is to make that scalar normalization compatible with this anisotropic paired family. The sign of r is unrestricted. Reference normalization is separated from the independent positive barrier coefficient α and quadratic coefficient β .

Proposition 2. *Equation (49) is objective, is polyconvex in two dimensions, and satisfies $W(\mathbf{I}) = 0$ and $\partial_e W(\mathbf{0}) = \mathbf{0}$. It grows without bound as $J \rightarrow 0^+$, as $J \rightarrow \infty$, and as $\|\mathbf{F}\| \rightarrow \infty$ at bounded positive J .*

Proof. The composition $H(\mathbf{z}(\mathbf{F}, J))$ is convex in independent $(\mathbf{F}, J)$ because each z_i is convex and H is convex and nondecreasing. The correction $-r(J - 1)$ is affine in J for either sign of r . Every remaining nonconstant term is convex in $(\mathbf{F}, J)$. In two dimensions, convex dependence on $(\mathbf{F}, J)$ supplies the required convex representation in deformation minors. Objectivity follows from the norms and determinant. Equation (47), together with $\partial_e J|_0 = (1, 1, 0)^T$, gives exact reference normalization. Finally, the supporting plane of H and $z_i \geq 0$ give a constant lower bound for H ; the positive logarithmic, quadratic, and $\|\mathbf{F}\|^2$ terms dominate the affine correction in the stated limits. $\square$

These are constitutive statements. They do not prove uniqueness of a structural equilibrium, absence of geometric buckling, or convergence of a particular Newton algorithm. Nor do they prove that a homogenized material with an arbitrary microscopic instability can be accurately represented by a globally polyconvex law.

4.6 ICNN and integrated-hat spline cores

The ICNN uses softplus activations and nonnegative hidden, input, and output weights. Convexity of ICNNs follows from the composition construction described by Amos et al. [34]. Nonnegativity of the direct-input weights is an additional restriction needed here for monotonicity in the nonlinear features; convexity alone does not require every direct-input weight to be nonnegative.

For the spline core, each scalar edge has the form

$$f(x) = ax + \sum_k c_k B_k(x) + d, \quad a, c_k \geq 0, \quad (50)$$

where B_k'' is a nonnegative triangular hat. With $s = (x - k)/h$,

$$B_k(x) = h^2 \begin{cases} 0, & s \leq -1, \\ (s + 1)^3/6, & -1 < s \leq 0, \\ (-s^3 + 3s^2 + 3s + 1)/6, & 0 < s < 1, \\ s, & s \geq 1. \end{cases} \quad (51)$$

Thus f' and f'' are nonnegative, and f'' is continuous. Positive sums and layer compositions preserve convexity and monotonicity. Linear tails match analytically. Unlike a positive cubic-hinge construction, the curvature can decrease back to zero without becoming negative. This is a local C^2 spline implementation inspired by the monotonic input-convex KAN framework [38]; it is not an unmodified reproduction of that paper's spline implementation or training procedure.

The distinction at the tails is mathematical rather than terminological. Appendix A.3 of Ref. [38] specifies endpoint slopes using finite secants. For a strictly convex interior curve, a forward secant at the left endpoint is larger than the right derivative there, so that using it as a left-tail slope creates a downward derivative

jump. An analogous issue occurs at the right endpoint. Exact tangent slopes would preserve convexity, but a finite-secant formula does not generally guarantee exact convexity or C^1 continuity across the joins. This observation concerns extrapolation beyond the natural spline interval; it does not negate the interior constrained-spline argument. Equation (51) avoids this distinction through exact matching, with vanishing curvature at both tails.

4.7 A separate nonnegative-energy check

Zero reference energy and stress do not, on their own, establish global nonnegative energy. This distinction is also explicit in Linden et al. [9]. For Eq. (49), define $k_i = 1/p_i + 1/q_i$ and $\eta_i = \rho_i/k_i$. The determinant inequality $\|\mathbf{F}\mathbf{d}_i\|^2\|\mathbf{F}\mathbf{d}_i^\perp\|^2 \geq J^2$ gives $z_i \geq k_i J^{\eta_i}$. Together with $\text{tr } \mathbf{C}/2 \geq J$ and the supporting plane of H , this yields

$$W \geq g(J) := \sum_i h_i k_i [J^{\eta_i} - 1 - \eta_i(J - 1)] + (\alpha + \varepsilon)(J - 1 - \log J) + \frac{\beta}{2}(J - 1)^2. \quad (52)$$

Since $g(1) = g'(1) = 0$, proving $g'' \geq 0$ for $J > 0$ is sufficient. Appendix B describes the implemented bound. This is a saved-parameter certificate with numerical margins, not a restriction enforced at every training iterate and not a formally verified interval-arithmetic proof.

5 Numerical example I: constitutive response of a periodic porous cell

5.1 Purpose and configuration

This example asks whether a mechanically constrained learned law can represent the anisotropic response of the chosen cell, and which mechanical properties distinguish the learned alternatives. It does not measure structural speedups. The square periodic cell contains a centered elliptical void with area fraction 0.20, aspect ratio 2, and major-axis angle 30° relative to the cell axes; see Fig. 2. This orientation avoids the mirror-aligned configurations of the square cell. Centrosymmetry remains, but does not mean that the strain energy is an even function of strain.

The solid is compressible Neo-Hookean in plane strain:

$$\psi_\mu = \frac{\mu}{2}(\text{tr } \mathbf{C}_\mu - 2) - \mu \log J_\mu + \frac{\lambda}{2}(\log J_\mu)^2, \quad (53)$$

with Young’s modulus 1.628 GPa and Poisson ratio 0.4. Here μ and λ are the usual Lamé constants. This is the specific matrix law used by the reference solver, not a generic assertion about every Neo-Hookean variant. The production mesh contains 1546 quadratic triangles, 3240 nodes, and 4638 quadrature points. Periodicity and null-space constraints leave 6320 independent microscopic displacement coordinates. Cell averages use the complete reference cell measure.

The production mesh was selected using a local stress-field criterion along an investigated uniaxial path, rather than only a homogenized average. At $E_{11} = 0.20$, its maximum microscopic first Piola stress norm differs by approximately 0.486% from a 3095-element reference, whereas the average axial stress differs by approximately 0.001%. This is a mesh check at a specified state, not a uniform error bound over all multiaxial data.

5.2 Data domain and model selection

The fitting domain is informed by a preliminary macroscopic deformation study. The final training box, shared by the constitutive and reduced models, has the following rounded bounds:

$$\mathbf{e}_{\min} = (0, -0.09865, -0.16807)^T, \quad \mathbf{e}_{\max} = (0.1955, 0.01635, 0.10793)^T. \quad (54)$$

The new constitutive features do not change this box. For the selected polyconvex network runs, 4208 of the 4950 available training states are used for fitting and 742 for validation, with split seed 5. The independent test set contains 400 states. An additional 350-state probe extends selected distances from the box center by factors 1.05, 1.10, 1.25, 1.50, and 2.00. Five probe labels are nonfinite following failed reference solves and are excluded with the same finite-label mask for all compared models. Their exclusion is reported, not interpreted as proof of a physical instability.

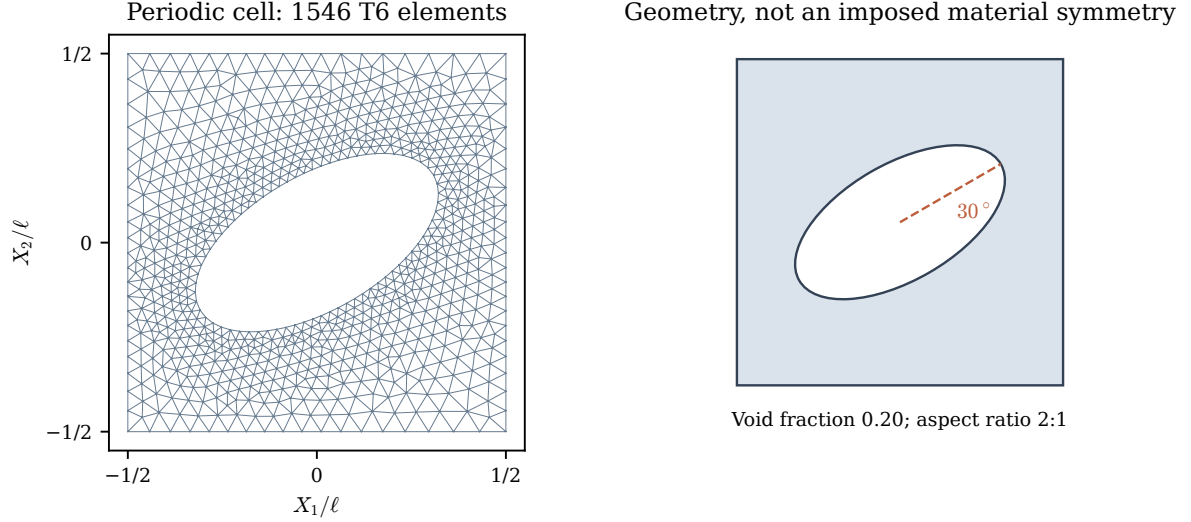


Figure 2: Microstructure and production discretization. The left panel shows corner-edge traces of the quadratic mesh; the right panel specifies the void geometry. The cell side is denoted by ℓ . No material reflection symmetry is imposed on the learned paired features.

The selected feature-learning runs use full-batch Adam for 2600 steps followed by up to 400 L-BFGS iterations. The normalized training objective combines stress, energy, and reference-tangent relative mean-square errors with weights 1, 0.2, and 0.02, respectively. The tangent target is obtained from a separate coarse periodic reference calculation. Candidates are selected by validation stress error; test and probe labels are not used in that second-stage selection. Both selected runs use initialization seed 16, with 32 features and hidden widths 24/24 for ICNN and 12/12 for ICKAN. Earlier baselines have different feature families, objectives, and training budgets. Comparisons to them diagnose practical approximability, not compute-matched superiority of one network architecture.

5.3 Reduced-state representation for this material

The 4950 displacement snapshots provide the basis of the reduced microscopic models as well as the effective constitutive labels. Figure 3 separates POD compression from the strain-informed coordinate fit. The relative snapshot truncation norm for the retained 39-dimensional space is 9.94×10^{-7} . After its orthogonal rotation into primary and secondary spaces, $n = 3$ and $\bar{n} = 36$ in the notation of Section 3.

The fitted relation between the deployed primary coordinates and macroscopic strain has an aggregate relative array discrepancy of 3.60×10^{-5} on these snapshots; its maximum absolute component discrepancy is 2.55×10^{-5} . These quantities explain why macroscopic strain is a useful primary-coordinate predictor for this material. They do not imply exact equality at unsampled states, nor do they measure the closure error or the equilibrium residual. In particular, the direct substitution in D-HPROM-ANN is an additional approximation, even when the plot appears visually indistinguishable from the identity.

The secondary-coordinate closure is a 3–64–64–36 feed-forward network with two tanh hidden layers and a linear output. Its training script uses input standardization, the raw secondary-coordinate loss in Eq. (20), and a 90/10 internal fit/validation split with seed 20260903. Full-batch Adam has initial learning rate 2×10^{-3} , a validation-driven reduction schedule, and a maximum budget of 20000 epochs. The archived best relative validation mean-square error is 3.75×10^{-7} . This is an internal closure metric after a POD built from all 4950 snapshots, not an independent material-generalization estimate. Independent effective-stress and structural tests provide the latter assessment.

Each deployed adaptive cubature field has architecture 3–128–128–128–10, GELU hidden activations, and the scaled softmax output of Eq. (30). The residual and stress checkpoints use distinct ten-element supports. Their larger neural cores illustrate why the online cost cannot be inferred solely from the three equilibrium coordinates or from the element count. These reduction-training choices differ from those of the constitutive networks; the common material and data domain do not imply matched offline training budgets.

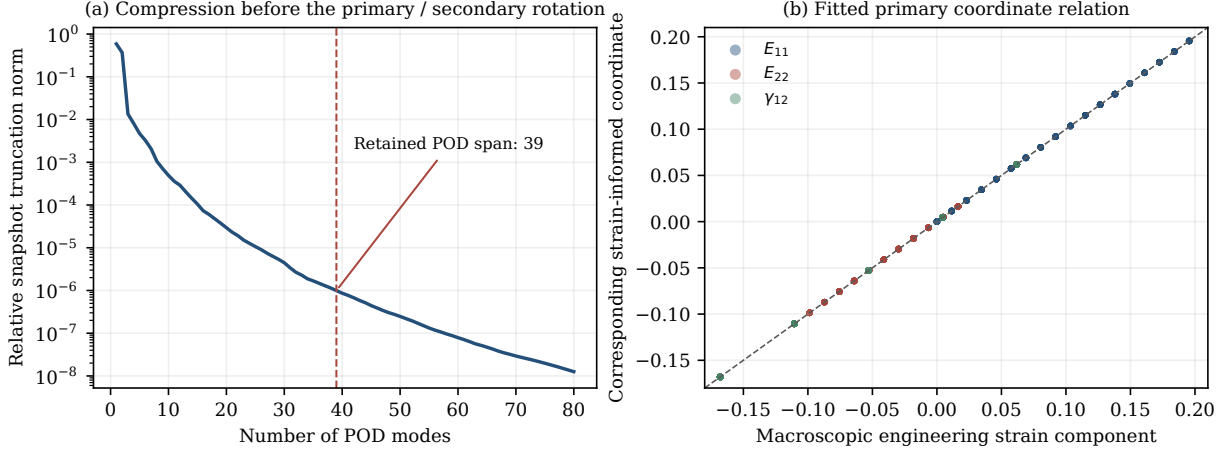


Figure 3: Diagnostics from the deployed 39-mode basis artifact. Left: relative POD snapshot truncation norm, before the primary/secondary rotation. Right: strain-informed primary coordinates versus the corresponding macroscopic strains; every tenth snapshot is plotted, whereas the reported discrepancies use all 4950 states. Neither diagnostic measures structural accuracy or proves validity outside the sampled domain.

5.4 From a restrictive feature family to an accurate fit

The earlier 15-feature construction combines $\text{tr } \mathbf{C}$, twelve balanced directional fourth- and sixth-power features of $\mathbf{F}$ and $\text{cof } \mathbf{F}$, and J, J^2 . Its reference correction uses $-r \log J$. The revised formulation changes both the feature family and the separation of reference normalization from volumetric stiffness; it is not simply a wider network or a relabelling of the old inputs.

A convex nondecreasing core preserves the componentwise ordering of its inputs. Consequently, if all features increase along a deformation path, the core output cannot decrease. Whether the complete energy decreases also depends on its analytic correction terms. An incompatibility with a fixed feature/correction family is a representational restriction of that family, not a contradiction of polyconvexity as a mathematical framework. The revised directional powers and determinant exponents provide additional ways to capture the observed deformation response while maintaining the sufficient convexity conditions.

The saved original ICNN and ICKAN have held-out aggregate stress errors of approximately 22.78% and 18.33%, respectively. The selected revised models reduce these errors to approximately 0.65%; Table 4 reports the independently audited selected checkpoints. The Free model’s corresponding aggregate stress errors are 1.2643% on the test set and 5.5024% on the finite probe set. Thus the polyconvex models improve this test fit but do not outperform Free on aggregate probe error. This distinction matters when discussing extrapolation.

Table 4: Selected polyconvex models: constitutive errors, in percent.

Model	e_S^{test}	e_W^{test}	e_S^{probe}	Worst test	Worst probe
ICNN	0.6491	0.2169	7.0582	5.53	43.60
ICKAN	0.6551	0.2148	7.0227	5.78	45.92

All values are percentages. The last two columns are maximum per-state stress errors; the first three are aggregate array norms. Test: 400 states; probe: 345 finite reference states out of 350.

The archived independent audit records successful energy first- and second-derivative checks and saved-weight nonnegative-energy sufficient bounds for both selected models. The conservative finite-interval margins are 17.25 and 16.82 in normalized energy units. Random admissible-deformation tests support implementation consistency, but the polyconvexity argument is analytical. Neither aggregate error nor a successful bound on energy is a uniform bound on stress prediction error.

5.5 A closed-cycle witness for direct stress regression

For a differentiable hyperelastic potential, any closed strain cycle in its domain satisfies

$$\mathcal{W}_{\text{cyc}} = \oint \mathbf{s} \cdot d\mathbf{e} = 0. \quad (55)$$

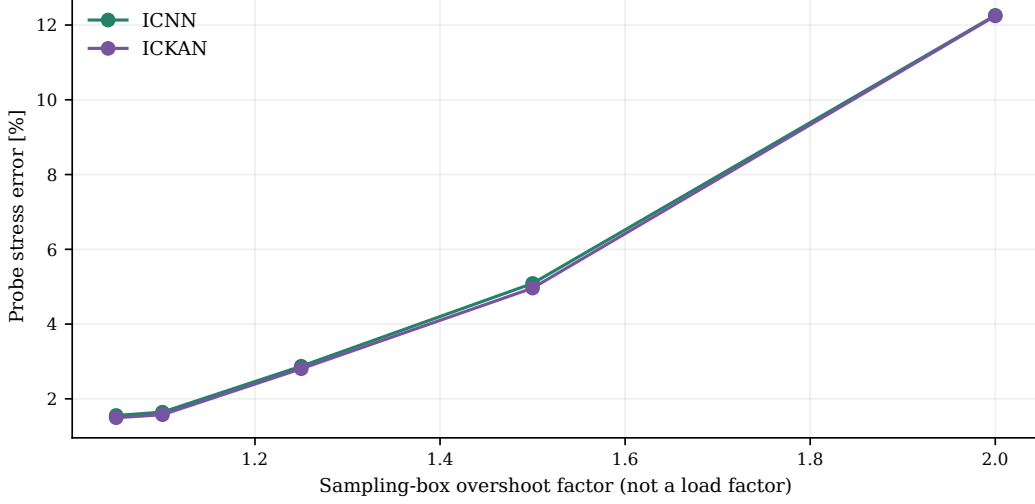


Figure 4: Stress error on increasingly distant probe rings, using only finite reference labels. Ring factors expand the sampling box; they are not coupon load multipliers. The increase to approximately 12.25% at factor 2 illustrates a remaining extrapolation limitation despite architectural admissibility.

This test isolates integrability and does not require a structural Newton failure. The importance of conservative strain–stress mappings is discussed explicitly by As’ad et al. [7]. We evaluate a counterclockwise rectangular cycle in (E_{11}, E_{22}) , with half-width 0.001, centered at

$$\mathbf{e}_c = (0.0051260033, -0.0945507613, -0.1283720602)^T. \quad (56)$$

The engineering shear component is held fixed. The rectangle lies inside Eq. (54). Its center was chosen after inspecting regression tangent asymmetry at held-out states; it is a deliberately selected mechanical witness, not an unbiased estimate of the frequency of violations. The models remain frozen during this audit.

Table 5 retains the signed work and its quadrature dependence. Reversing the cycle reverses the regression work, which is incompatible with interpreting that map as an elastic potential. The result establishes a constitutive inconsistency, not by itself a quantitatively important error for every macroscopic loading history. The Free model passes this test because it is energy-based; energy-cycle consistency therefore cannot distinguish Free from the polyconvex models.

Table 5: Closed-cycle work under quadrature refinement.

Points/edge	Regression	Free	ICNN	ICKAN
8	-213.922	-5.82077e-11	0	-0.0215596
16	-213.922	1.89175e-10	-6.36646e-11	0.00629948
32	-213.922	2.04182e-10	6.86668e-11	-0.000200597
64	-213.922	-5.10454e-11	-6.5711e-11	-5.25639e-05
128	-213.922	2.33996e-10	3.18607e-10	4.62179e-06

Signed cycle work in J m^{-3} . FOM at 8 points per edge: $-6.98492\text{e-}10 \text{ J m}^{-3}$. The spline quadrature error decreases with refinement; it is not physical dissipation.

5.6 Rank-one curvature: guarantees, samples, and an unresolved reference

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, the relevant finite-strain curvature is

$$D_{\mathbf{F}}^2 W[\mathbf{H}, \mathbf{H}] = (\delta \mathbf{e})^T \mathbf{D} \delta \mathbf{e} + \mathbf{S} : (\mathbf{H}^T \mathbf{H}), \quad \delta \mathbf{E} = \text{sym}(\mathbf{F}^T \mathbf{H}), \quad (57)$$

where $\delta \mathbf{e}$ uses Eq. (2). Omitting the second term tests a different property. In particular, a positive semidefinite tangent with respect to $\mathbf{e}$ does not generally imply rank-one convexity in compression. For energy-based models, Eq. (57) is evaluated over sampled states and unit directions; negative candidates are checked by second differences of $W(\mathbf{F} \pm h\mathbf{H})$.

No negative direction is found for Free in the audited test, finite-label probe, and additional in-box sets. An outward search over 1800 strain-space rays finds its first sampled negative value at expansion factor 1.55,

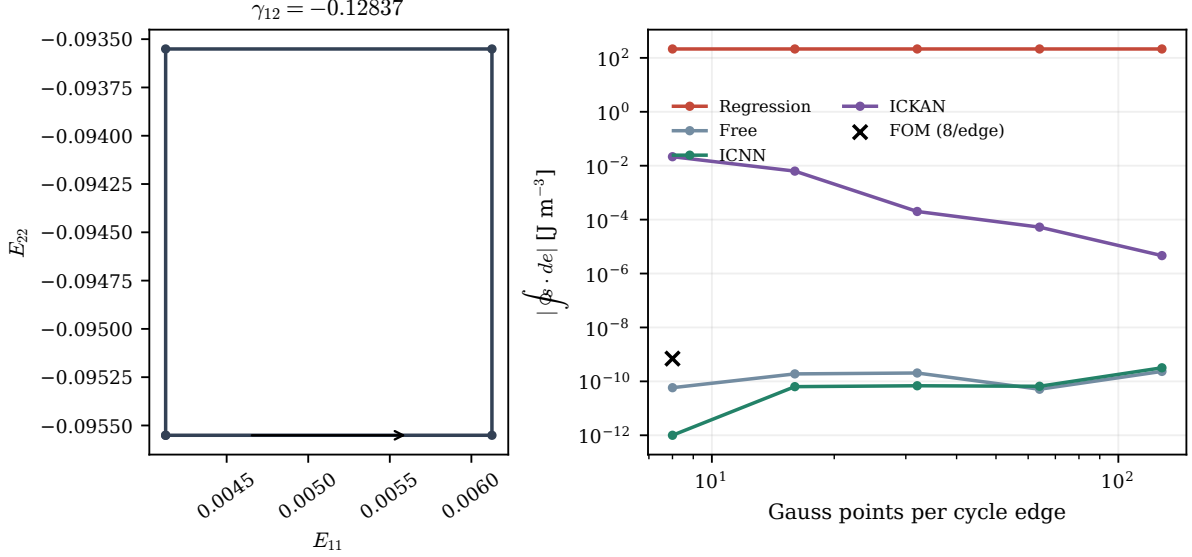


Figure 5: Closed-cycle geometry and quadrature refinement of net stress work. Regression approaches a nonzero signed value of $-213.922 \text{ J m}^{-3}$. Free and ICNN cancel to numerical precision; an exact numerical zero is plotted at $10^{-12} \text{ J m}^{-3}$ solely for logarithmic display. The C^2 spline energy also defines a conservative law, but crossing spline knots requires quadrature refinement; its residual work decreases towards zero. The FOM marker is a separately evaluated cycle at eight Gauss points per edge.

Table 6: Finite rank-one curvature audit. Positive sampled values do not establish a global property.

State set	N	Free	ICNN	ICKAN
Held-out test	400	176.681	220.486	220.468
Finite-label probe	345	113.034	203.280	203.538
In-box audit (unlabelled)	12000	163.040	213.979	211.168

Minimum sampled rank-one curvature in MPa. Minima need not occur at the same state or direction. The additional in-box cloud has no independently computed FOM labels; 180 unit directions for $\mathbf{b}$ are sampled, with minimization over $\mathbf{a}$ via an eigenproblem. None of these finite samples is a global certificate.

with $J \simeq 0.780$ and curvature approximately -3.43 MPa. This is a result of that search, not the exact nearest violation. Independent energy finite differences confirm the negative curvature of the learned model.

However, the attempted FOM continuation fails numerically before reaching that candidate. No refined-continuation or global branch-tracking result is available there. Therefore this candidate establishes that the Free network lacks global rank-one convexity, but does not establish disagreement with a converged stable homogenized reference at that state. Solver nonconvergence is not evidence that the physical branch ceases to exist. The stronger claim of an FOM-confirmed false instability is left open. The ICNN and ICKAN have an analytical polyconvexity argument; their sampling audit is a check of implementation, not its replacement.

6 Numerical example II: structural accuracy and online cost

6.1 Purpose, boundary conditions, and coverage

This example asks how the constitutive approximations affect a complete nonlinear structural calculation. The material is the same periodic cell as in Section 5; the new evidence concerns error propagation and online cost, not a second microstructure. The dog-bone profile has overall length 165 mm, grip width 19 mm, narrow-section width 13 mm, narrow-section length 57 mm, and fillet radius 76 mm. It is a plane-strain computational benchmark using an ASTM-inspired in-plane profile, not a simulation certified to reproduce an ASTM tensile test.

The two end faces receive equal-and-opposite axial resultants of 100 kN per end, applied in 20 equal increments. Three point constraints remove rigid modes: the axial and transverse displacement at the left mid-edge node, and transverse displacement at the right mid-edge node. Entire end faces are not clamped. Consequently, a small skew of the deformed ends is not ruled out by the boundary conditions. The anisotropic response can couple axial extension to shear.

The macro mesh contains 690 quadratic triangles, 1501 nodes, and 2070 integration points. The reference extrusion thickness used to convert the two-dimensional stress integrals into the stated force resultants is 50 mm. This force convention is part of the plane-strain benchmark, not an ASTM specimen-thickness specification. The physical cell size is declared as 1.3 mm, corresponding to ten cells across the narrow width. This provides a scale-separation assumption, not a direct numerical validation of homogenization error at that ratio. All saved converged integration-point strains for every reported model remain inside the unchanged training box. This check does not automatically cover unsaved intermediate Newton iterates or certify another load case.

6.2 Online models and implementation verification

The linear HPRM uses $n_{\text{tra}} = 39$ generalized coordinates, a residual rule supported on 135 elements, and a stress rule on 73 elements; their union contains 183 elements. The iterative HPRM-ANN uses $n = 3$ primary generalized coordinates and $\tilde{n} = 36$ secondary generalized coordinates supplied by the closure map, with ten residual and ten stress elements and a union of 19. The direct model evaluates stress on ten elements and solves no microscopic equilibrium equations. These counts refer to actual reduced MDPA meshes, not masked evaluations on the full RVE mesh. Figure 7 shows their spatial distribution and the overlap of residual and output rules.

For the optimized linear and direct implementations, the maximum normalized stress differences from the original implementations over the verification states are 4.71×10^{-14} and 4.20×10^{-13} . Corresponding tangent differences are 2.68×10^{-10} and 7.25×10^{-10} . Independent finite-difference checks give discrepancies of approximately 3.51×10^{-9} and 2.38×10^{-9} . Such comparisons check that optimization preserves the deployed model; they do not establish that either model exactly reproduces full microscopic equilibrium.

6.3 Error and timing definitions

For a saved nodal or integration-point array A , the reported relative error is

$$e_A = 100 \frac{\|\text{vec}(A - A_0)\|_2}{\|\text{vec}(A_0)\|_2}, \quad (58)$$

where A_0 is the nested reference on the same meshes. These are unweighted discrete array norms. The strain array stores engineering shear, whereas the stress array stores tensor shear. They are neither quadrature-weighted continuum L^2 norms nor invariant tensor Frobenius norms. The metric is kept identical across models; this does not make the three different field errors interchangeable.

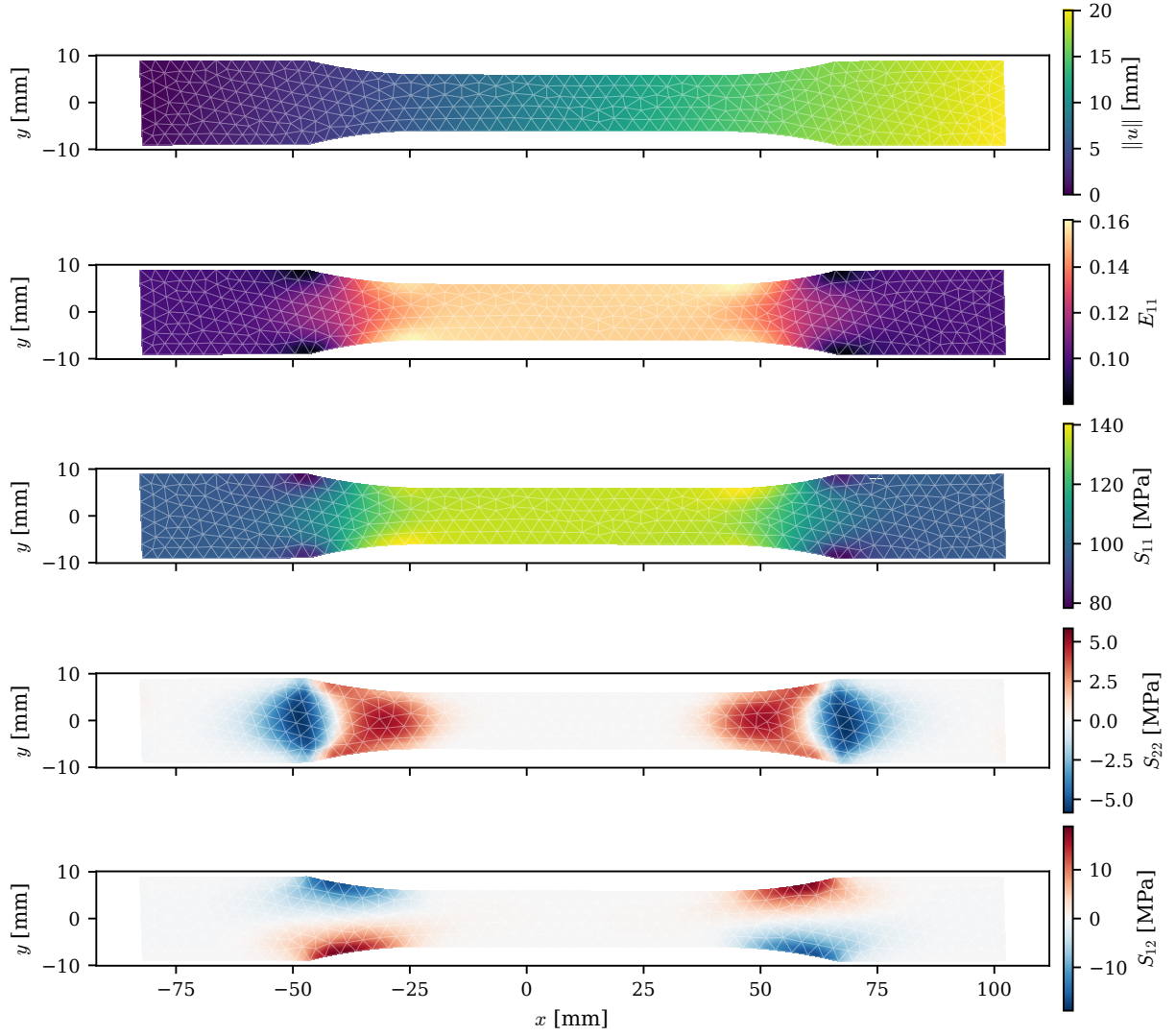


Figure 6: Nested FOM–FE² reference at 100 kN per end. Fields are displayed on the actual deformed mesh, without displacement amplification. Gauss-point strain and stress values are first averaged within each element and then patch-averaged to nodes for visualization only. These recovered contours are not the raw fields used in error norms and can smooth local extrema. Stresses are second Piola components. A three-dimensional von Mises stress is not reconstructed because the saved output does not include the necessary out-of-plane Cauchy stress.

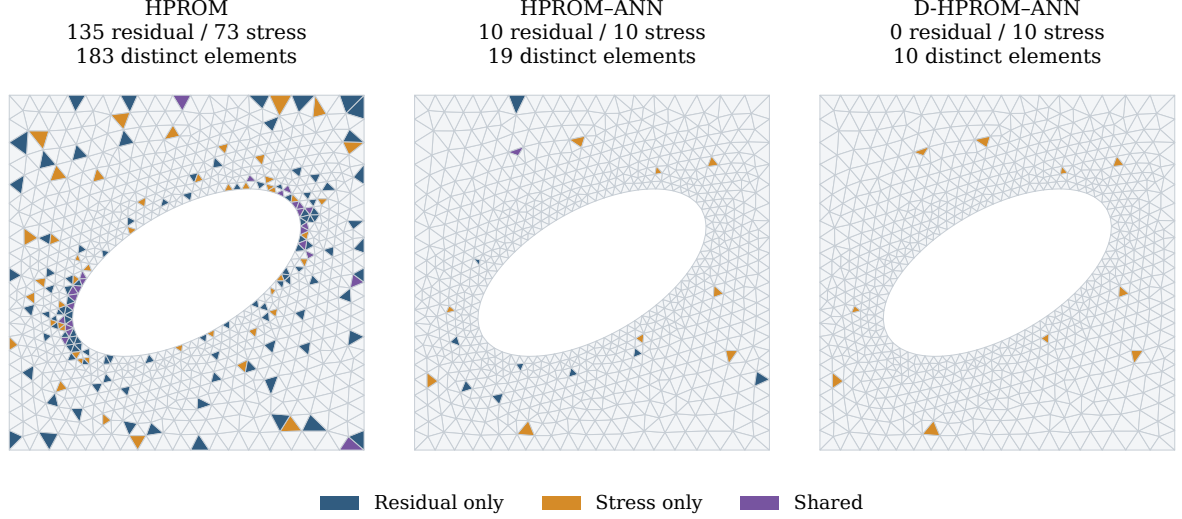


Figure 7: Element supports of the deployed hyperreduced microscopic models. The pale full-cell mesh provides location context only; discarded elements are not assembled online. Colors distinguish residual-only, stress-only, and shared elements. Supports are read from the same index arrays used to construct the reduced MDPA meshes.

Online wall time includes the complete macroscopic solve, microscopic or surrogate stress/tangent evaluations, communication, and lazy worker construction within the timed solve. Parent setup, offline data generation, training, reduced-mesh construction outside the timed region, and result export are excluded. The FOM uses 20 worker processes with one numerical-library thread per worker. The three optimized microscopic surrogates use the same process configuration and three sequential fresh complete runs. PANN evaluations are batched in one process and use the separately investigated thread settings shown in Table 7.

The macroscopic solve stops when the free-degree-of-freedom residual norm falls below 10^{-5} N or its ratio to the initial increment residual falls below 10^{-7} . The periodic FOM uses a relative microscopic displacement-increment tolerance of 10^{-10} , with continuation when needed. The archived FOM and selected PANN times each have one repetition in this comparison. Thus their ratios are observed online speedups under the stated implementations, not matched-thread efficiency estimates or repeated-run confidence intervals. Table 8 exposes the available reduced-model variability rather than selecting its fastest repetition. A publication-final timing study should repeat every tier under a common resource-budget protocol and retain all repetitions.

Table 7: Complete coupon simulations: final-field errors against the nested FOM and archived online wall times.

Model	CPU	n_{run}	t [s]	t_0/t	e_u [%]	e_E [%]	e_S [%]
FOM	20p	1	4268.679	1.0	0.00000	0.00000	0.00000
HPROM	20p	3	67.348	63.4	0.00565	0.01653	0.00099
HPROM-ANN	20p	3	30.143	141.6	0.05200	0.14968	0.03727
D-HPROM-ANN	20p	3	3.788	1126.9	0.02324	0.04851	0.00967
ICNN	4t	1	1.734	2461.3	0.42468	0.77028	0.08880
ICKAN	8t	1	3.669	1163.5	0.38657	0.78932	0.07800
Free	12t	1	1.356	3146.9	0.92988	1.49464	0.13492
Regression	16t	1	0.946	4510.4	0.60620	0.81218	0.07378

p: worker processes, one numerical-library thread each; t: PyTorch threads. For $n_{\text{run}} = 3$, time is the median; for $n_{\text{run}} = 1$, it is the archived single run. These are online wall times, not offline-inclusive or matched-thread speedups.

Table 8: Sequential optimized-run repetitions used for median HPROM timings.

Model	t_1 [s]	t_2 [s]	t_3 [s]
HPROM	63.354	67.348	68.953
HPROM-ANN	27.947	34.628	30.143
D-HPROM-ANN	3.788	3.758	3.969

6.4 Accuracy–cost trade-off and Newton behavior

Figure 8 and Table 7 show distinct operating points rather than a single universally best model. The linear HPROM gives the smallest displacement error, approximately 0.00565%, at a median online time of 67.35 s. The iterative HPROM–ANN reduces this time to 30.14 s with a displacement error of approximately 0.0520%. The optimized iterative nonlinear model is therefore about 2.23 times faster than the *optimized* linear model; comparison against the old linear implementation would overstate this gain. The direct closure reduces the time to 3.79 s and has a displacement error of approximately 0.0232% on this path. Its lower error than the iterative nonlinear model is an observed outcome, not a consequence of a general error ordering.

The ICNN and ICKAN times are 1.73 s and 3.67 s, with displacement errors of approximately 0.425% and 0.387%. Their finite-strain structural accuracy is therefore consistent with a useful constitutive approximation, although it is poorer than the reduced microscopic models on this benchmark. Direct stress regression is fastest, but its successful coupon solution does not remove the integrability defect exposed by Section 5.5. Free also converges and remains a relevant baseline; its absence of a global convexity constraint is not equivalent to a failure on this load case.

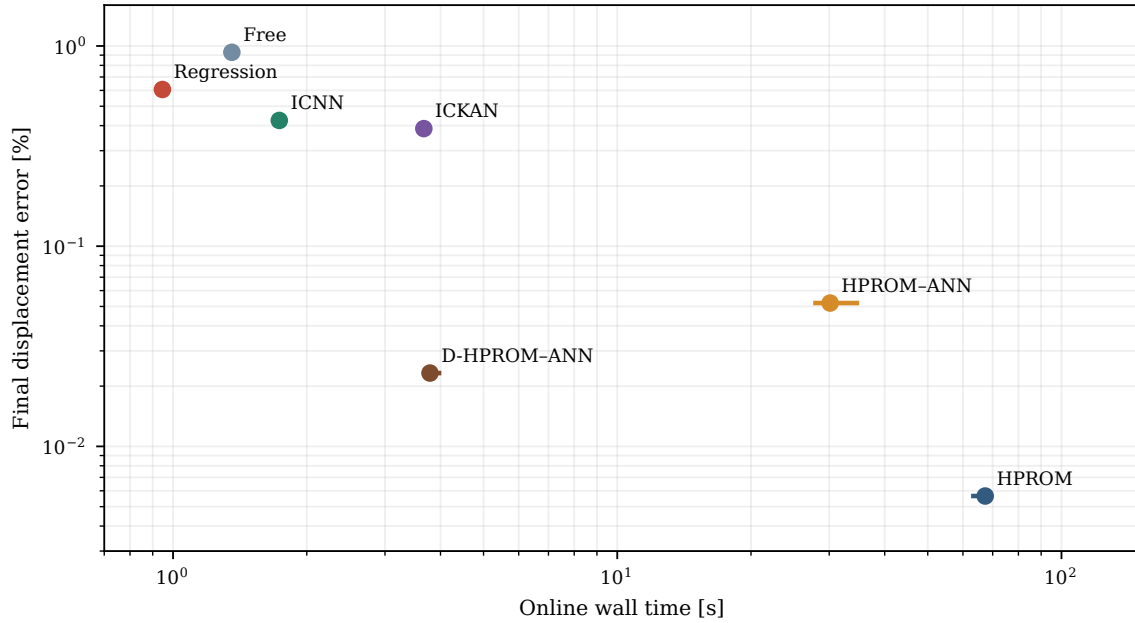


Figure 8: Online cost versus final displacement error on the common coupon problem. Horizontal segments show the observed range of three optimized HPROM runs; unbarred points are archived single PANN runs. Process and thread counts differ as detailed in Table 7. The plot does not include offline costs.

All models require four macroscopic iterations per increment in the archived runs. Consequently, this example provides no empirical evidence that polyconvexity reduced Newton iterations relative to Free or Regression. It establishes that the selected constrained models can be used successfully with the common structural solver, while maintaining their constitutive guarantees. A separate challenge problem would be needed to establish a solver-robustness advantage, and its reference solution would need to be validated independently.

7 Conclusions

Paired directional features with constrained trainable powers and determinant exponents provide a practical polyconvex constitutive construction for the anisotropic periodic material studied here. Their isotropic reference derivatives allow exact stress normalization through an affine determinant correction without tying the normalization coefficient to the volumetric barrier stiffness. Both a smooth ICNN and an integrated-hat spline-ICKAN achieve approximately 0.65% held-out aggregate stress error and below 0.43% displacement error in the nominal coupon simulation.

Optimized reduced microscopic models occupy a different accuracy–cost range. They are more accurate on this coupon, while direct learned constitutive evaluation is generally faster. A closed-cycle test exposes a conservative-mechanics limitation of the regression baseline that nominal finite element convergence does

not reveal. In contrast, the available Free-model curvature witness is not yet supported by a converged FOM reference and is reported only within that limited scope. The study supports the usefulness of the proposed constrained feature family for the present problem, not unrestricted extrapolation or a universal ranking of constitutive and reduced-order models.

A Joint convexity of the directional building block

Consider $f(\mathbf{x}, J) = \|\mathbf{x}\|^a J^{-b}$ with $J > 0$, $a \geq 1$ and $0 \leq b \leq a - 1$. For $t = \|\mathbf{x}\| > 0$, the Hessian block in the radial and determinant variables is

$$\begin{pmatrix} a(a-1)t^{a-2}J^{-b} & -abt^{a-1}J^{-b-1} \\ -abt^{a-1}J^{-b-1} & b(b+1)t^aJ^{-b-2} \end{pmatrix}. \quad (59)$$

Its diagonal entries are nonnegative and its determinant is

$$ab(a-b-1)t^{2a-2}J^{-2b-2} \geq 0. \quad (60)$$

The transverse Hessian eigenvalues are $at^{a-2}J^{-b} \geq 0$. Continuous extension gives convexity at $\mathbf{x} = 0$. For the constitutive evaluation on $\text{GL}^+(2)$, a nonzero reference direction cannot satisfy $\mathbf{F}\mathbf{d} = 0$, so this point does not introduce an interior singularity of the directional norm. Taking $a = 2p$ or $2q$ establishes Eq. (46).

B Sufficient lower bound for nonnegative energy

Write $t = \|\mathbf{F}\mathbf{d}_i\|^2$, $u = \|\mathbf{F}\mathbf{d}_i^\perp\|^2$. Hadamard's determinant inequality gives $tu \geq J^2$. Since each feature is increasing in t, u , its minimum at fixed J occurs on $tu = J^2$. Minimizing

$$\frac{t^{p_i}}{p_i J^{b_i}} + \frac{J^{2q_i - c_i} t^{-q_i}}{q_i} \quad (61)$$

gives $t^{p_i+q_i} = J^{2q_i+b_i-c_i}$ and hence $z_i \geq k_i J^{\eta_i}$. This is a separate lower bound for each feature; simultaneous attainability for all directions is not assumed. It therefore remains valid after summing the supporting-plane inequalities.

For Eq. (52),

$$J^2 g''(J) = \alpha + \varepsilon + \beta J^2 + \sum_i c_i^* J^{\eta_i}, \quad c_i^* = h_i k_i \eta_i (\eta_i - 1). \quad (62)$$

Only terms with $0 < \eta_i < 1$ can have negative coefficients. If $c_- = \sum_{c_i^* < 0} |c_i^*|$, the two inequalities

$$\alpha + \varepsilon + \sum_{\eta_i < 0} c_i^* - c_- \geq 0, \quad (63)$$

$$\beta + \sum_{\eta_i \geq 1} c_i^* - c_- \geq 0 \quad (64)$$

are sufficient on $0 < J \leq 1$ and $J \geq 1$, respectively. For the first region, $J^{\eta_i} \leq 1$ for negative-coefficient terms and $J^{\eta_i} \geq 1$ when $\eta_i < 0$. For the second region, divide $J^2 g''$ by J : all negative-coefficient powers then satisfy $J^{\eta_i-1} \leq 1$, while $J^{\eta_i-1} \geq 1$ for $\eta_i \geq 1$ and $\beta J \geq \beta$. Dropping the remaining nonnegative contributions gives the stated bounds. Their failure is inconclusive.

The selected parameters require a sharper check. The audit partitions $[10^{-4}, 10^4]$ into 4000 logarithmically spaced intervals. On each interval $[J_L, J_R]$, a positive power term is bounded below by the smaller endpoint value, whereas a negative-coefficient term is bounded below by $c_i^* J_R^{\eta_i}$, since its exponent lies in $(0, 1)$. The constant and quadratic terms are bounded below by $\alpha + \varepsilon$ and βJ_L^2 . Below the finite interval, the nonnegative negative-exponent terms increase and the magnitudes of the negative-coefficient terms decrease toward $J = 0$. Above it, factoring out J^{η_-} with $\eta_- = \max_{c_i^* < 0} \eta_i < 1$ yields a lower bound whose retained positive powers increase. These endpoint and tail inequalities cover every $J > 0$, not only the grid points. Positive margins establish the sufficient analytical argument for the saved parameters to the accuracy of its floating-point evaluation; random energy samples alone would not establish it.

C Reproducibility and draft status

The accompanying `build_evidence.py` reads archived result arrays and metadata, verifies common mesh hashes, loads, completed increments, and saved-state coverage, and regenerates the numerical tables and vector figures. The generated `evidence_manifest.json` records SHA-256 hashes of the consumed artifacts. The additional `build_method_figures.py` generates the model-hierarchy diagram and computes the POD/primary-coordinate diagnostics from the retained basis; its separate manifest records that source hash. Neither script performs retraining or a new timing run. Selected checkpoint hashes and training manifests are retained in the constitutive audit directory. Thus regenerated graphics cannot silently be interpreted as newly measured performance.

The source text of the earlier manuscript is preserved. This draft changes its organization and narrows claims to available evidence. Its accompanying `REFERENCE_AUDIT.md` distinguishes extraction, inspected source passages, and unfinished full-paper reading. The supplied Boehler chapter is incomplete, and no theorem is attributed to its missing pages. Author metadata, complete reference review, final statistical timing protocol, and any additional numerical example remain editorial or experimental completion tasks rather than implicit completed results.

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